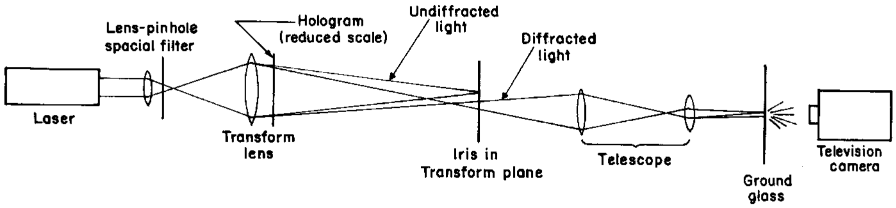


FIGURE 12-5
Optical reconstruction system for coded aperture imaging. (From Farmelant, M., et al.[3] Used with permission.)

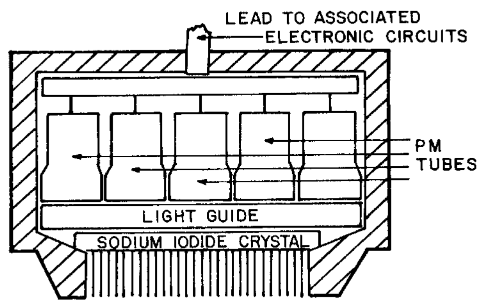
Many coded apertures have been studied, including some that change continuously over the time in which radiation from the patient is detected. All coded apertures developed so far have one major limitation, however; they exhibit very low sensitivity to large areas of radioactivity such as those usually encountered in nuclear medicine. Consequently, coded apertures have not found useful applications in clinical nuclear medicine.

Scintillation Crystal

Although NaI(Tl) crystals up to 0.5 m in diameter are used in single-crystal scintillation cameras, the periphery of the crystal is always masked and does not contribute to image formation. With a 28-cm-diameter crystal, for example, only the center 25 cm is used to create an image. Masking is necessary to prevent distortion in the image caused by inefficient collection of light at the periphery of the crystal. To prevent physical damage to the crystal and to keep it free of moisture, the crystal is sealed between a thin (<1 mm) aluminum canister on the exterior surface and a glass or plastic plate on the interior surface.

The photopeak detection efficiency of a 1.25-cm-thick NaI(Tl) detector varies from about 80% for ^{99m}Tc (140 keV) to about 10% at 511 keV. Increased detection efficiency could be achieved with thicker crystals. However, multiple scattering within the crystal would also increase and cause a reduction in the photon-positioning accuracy of the camera.

Although NaI(T1) is the detector used today in almost all clinical scintillation cameras, other detectors have been or are being explored. Among these detectors are bismuth germanate [$Bi_4Ge_3O_{12}$, often called BGO], barium fluoride [BaF_2], lutetium orthosilicate [$Lu_2[SiO_4]O(Ce)$, often called LSO], yttrium orthosilicate [YSO], and lutetium orthophosphate (LOP).



MARGIN FIGURE 12-5
Detector assembly of a scintillation camera.

Light Guide

In addition to the protective glass plate, camera detectors may have a light guide (usually composed of Plexiglas) that transmits and distributes the light from the primary interaction site in the crystal to the surrounding PM tubes. The distribution of light within a light guide of properly chosen thickness (usually about 4 cm from the central plane of the crystal to the face of the PM tubes) results in increased counting efficiency and positioning accuracy. The light is transmitted with almost no loss because the sides of the light guide are coated with a reflective covering. Because the light guide is solid, the detector assembly can be positioned in different orientations without difficulty. A light guide is not required in digital scintillation cameras.

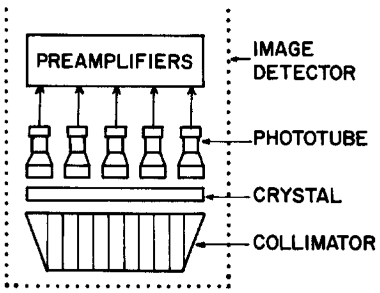
Photomultiplier Tubes

The light transmitted to a particular PM tube strikes the photocathode element of the tube. In response, electrons are ejected from the photocathode and focused upon the first of 8 to 14 positively charged electrodes referred to as *dynodes*. The number of electrons is multiplied in the remaining dynodes to yield 10^6 to 10^8 electrons collected at the final electrode or anode for each electron released at the photocathode. The actual number of electrons finally collected at the anode (i.e., the size of the output signal) varies in direct proportion to the amount of light striking the photocathode.

The photocathode of a PM tube is usually an alloy of cesium (Cs) and antimony (Sb) that has a relatively low *work function* (i.e., electrons are easily ejected by impinging light photons).

This lower limit of 70 keV, together with the upper limit of 600 keV, define the range of useful γ-ray energies for nuclear imaging.

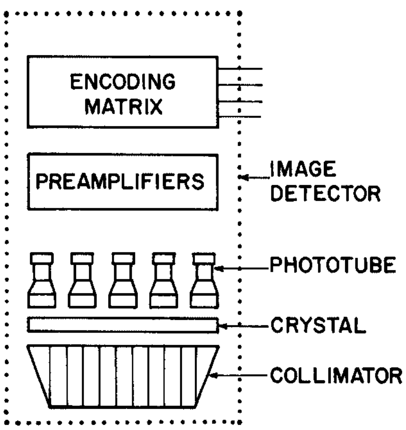
With a single-crystal scintillation camera, one of the restrictions on image quality is statistical fluctuation in the number of electrons ejected from the photocathode for a given amount of incident light. The statistical fluctuation is determined primarily by the average number of electrons ejected per unit of incident light. It has been improved markedly by replacement of older S11-type PM tubes with tubes having photocathodes that release more electrons per unit of incident light. Another improvement in noise reduction is the use of threshold preamplifiers that reject small signals from PM tubes distant from the interaction site in the crystal. These signals are rejected because they add significant noise without contributing substantially to image formation. Even with the newer tubes, however, statistical fluctuations in electron ejection from PM tube photocathodes establish a lower limit of about 70 keV for the energy range of γ rays amenable to single-crystal scintillation camera imaging. Also contributing to this lower energy limit is the difficulty in using pulse height analysis to distinguish primary from scatter photons when the primary photon energies are low. This difficulty arises because γ rays scattered during Compton interactions have about the same energy as the primary γ rays. Finally, low-energy γ rays are strongly absorbed in tissue, and few can reach the external detector.



MARGIN FIGURE 12-6
Detector assembly and preamplifiers of a scintillation camera.

Photomultiplier Tube Preamplifiers

Each PM tube transmits its output electrical signal to its own preamplier. This circuit is used primarily to match the PM tube impedance to that of the succeeding electronic circuits. Preamplifiers can be "tuned" so that signals from the PM tubes are identical for a given amount of energy absorbed in corresponding regions of the detector. Many scintillation cameras use this approach to tuning the PM tube array.

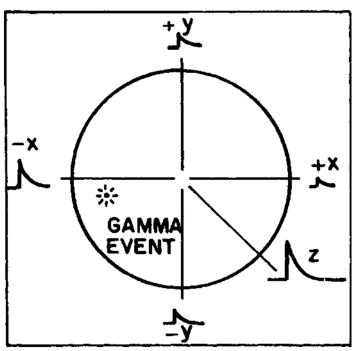


MARGIN FIGURE 12-7
Preamplifiers and position-encoding matrix of the scintillation camera.

Position-Encoding Matrix

The PM tubes are positioned in a hexagonal array above the scintillation crystal, with the spacing and separation from the crystal selected to provide an optimum combination of spatial uniformity and resolution. For the distribution of electrical signals emerging from the preamplifiers of these PM tubes, some operation must be performed to determine the site of energy absorption in the crystal. This operation is the function of the position-encoding matrix. The signal from each preamplifier enters the matrix and is applied across four precision resisters to produce four separate electrical signals for each PM tube. These signals are labeled $+x$, $-x$, $+y$, and $-y$.

For the PM tube in the center of the array, all four resistors are equal, and all four signals are identical. On the other hand, a PM tube in the lower left quadrant in the array is represented in the matrix by $+x$ and $+y$ resistors that are significantly different from the $-x$ and $-y$ resistors. A signal from the preamplifier of this PM tube furnishes $-x$ and $-y$ signals that are larger than the $+x$ and $+y$ signals. The actual magnitude of the signals from each four-resistor component of the encoding matrix reflects the proximity of the corresponding PM tube to the origin of the light flash within the crystal.



MARGIN FIGURE 12-8
In the position-encoding matrix for each PM tube, the PM signal is separated into four signals of relative sizes that reflect the position of the particular PM tube within the PM tube array.

Summation Amplifiers

From the encoding matrix, the position signals for each PM tube are transmitted to four summation amplifiers. The $+x$ signals for all of the PM tubes are added in one amplifier, the $-x$ signals for all of the tubes are added in a second summation amplifier, and so forth. The product of the summation operation is four electrical signals, one for each of the four deflection plates in the cathode-ray tube (CRT).

Deflection Amplifiers

Before the signals are applied to the deflection plates, they are transmitted to four deflection amplifiers where they are amplified, differentiated, and shaped (Figure 12-6).

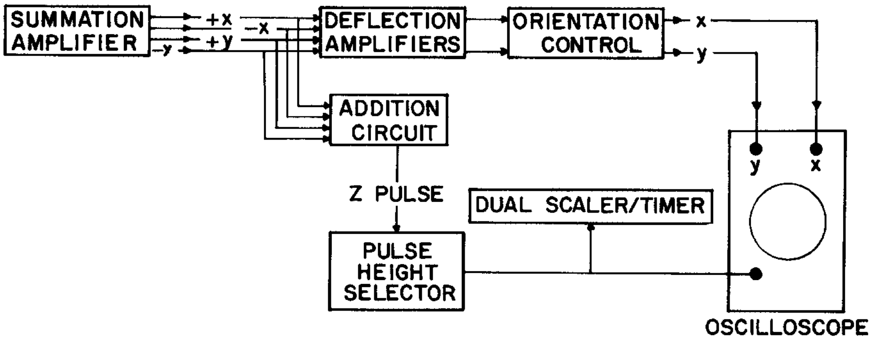


FIGURE 12-6
Position and pulse height control circuitry of a scintillation camera.

Here the signals also are divided by the Z pulse (see below) to normalize the positioning signals to the summation signal representing the particular interaction. This normalization process permits the use of a relatively wide pulse-size acceptance range (i.e., a relatively wide window) without an unacceptable reduction in position accuracy. Among other adjustments in the deflection amplifiers, the signals are modified to furnish the exact size of image desired on the CRT. Malfunctions of the deflection amplifiers often are responsible for distortions of the image such as the production of elliptical images for circular distributions of radioactivity. Following the deflection amplifiers, the positioning signals are fed to the orientation control where rotation switches on the camera console can be manipulated to alter the left–right and top–bottom presentation of the image on the camera console.

Pulse Summation (Addition) Circuit

In a circuit parallel to the deflection amplifiers, the four position signals are added to form the Z pulse. The Z pulse is sent to the pulse height analyzer for analysis of pulse height. In many cameras the pulse height analyzer is a fixed bias circuit, and the pulse height selection window is nonvariable. With a fixed-bias analyzer, pulses must be attenuated with an isotope range switch so that pulses representing total absorption of photons of a desired energy will be of the proper size to be accepted by the analyzer. Usually the isotope range switch, with different attenuation settings for different photon energy ranges, is positioned in the camera circuitry before the positioning signals branch to the addition circuit and deflection amplifiers.

If a particular pulse triggers the lower, but not the upper, discriminator of the analyzer, a negative pulse is transmitted to one input of an AND gate. The second input to the AND gate is furnished as soon as any previous pulse has dissipated. When the AND gate is activated, a microsecond unblanking pulse is furnished to the display CRT, where it activates the CRT electron beam to produce a short burst of electrons. This burst of electrons produces a tiny light flash on the cathode-ray screen. The number of electrons in the burst, and hence the brightness of the light flash on the cathode-ray screen, is adjusted with the intensity control on the console of the scintillation camera. The unblanking pulse is also transmitted to a scaler that records the counts forming the image.

Cathode-Ray Tube

The CRT consists essentially of an electron gun, four deflection plates, and a fluorescent screen that furnishes a flash of light when struck by the electron beam from the electron gun (Figure 12-7). Normally, the electron beam is prevented from flowing across the CRT. Only during the brief interval when the signal from the AND gate is received at the CRT is the electron beam permitted to flow and produce a light flash on the screen. During this brief interval, the position signals are applied to the

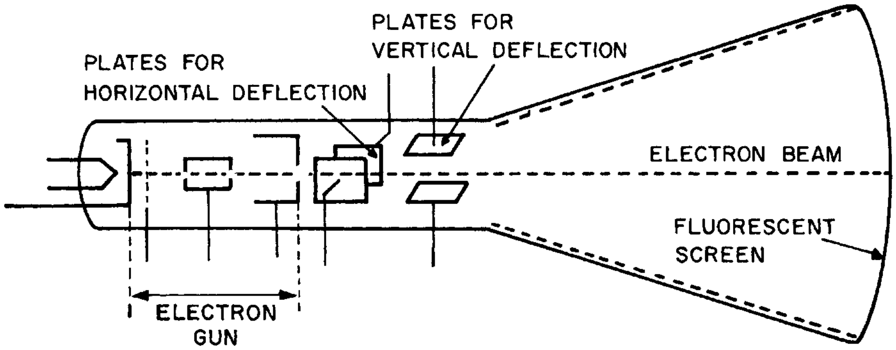


FIGURE 12-7
Cathode-ray image display device.

deflection plates so that the light flashes on the screen at a position corresponding to the site of γ-ray absorption in the crystal.

Scintillation cameras that digitize the signals after pulse-height and positioning analysis are called *hybrid analog-digital cameras*. Scintillation cameras that digitize the signals before pulse-height and positioning analysis are called *all-digital cameras*. Virtually all scintillation cameras marketed today are all-digital cameras. These cameras furnish more precise pulse-height and positioning analysis, and yield images with superior spatial resolution.

Processing, Display, and Recording

The image of a radionuclide distribution created in the scintillation camera may be digitized, stored in computer memory, and processed to enhance its features. The analog x and y signals from the camera electronics are converted to digital signals with analog-to-digital (ADC) converters. The digitized x and y signals determine the location of a pixel, and the number of counts recorded at that position determine the pixel value. Thus, images are stored as matrices in the computer in a special image display memory. The display system contains a video processor that reads the computer image display memory and performs a digital-to-analog conversion (DAC) to send the information to the monitor at video frame rates. A permanent record of the image may be made by exposing Polaroid or x-ray film to the light on the monitor. Recording devices with scanning laser beams may also be used to print images directly onto film.

Resolving Time Characteristics of Scintillation Cameras

At relatively low sample activities, the count rate from a single-crystal scintillation camera varies linearly with the sample activity. At higher sample activities, however, this relationship is not maintained, and the count rate increases less rapidly than expected as the activity of the sample is increased. The resulting loss of counts is termed the coincidence loss, and its magnitude depends on the resolving time of the camera. As described in Chapter 9, the resolving time is the minimum time required between successive interactions in the scintillation camera crystal for the interactions to be recorded as independent events.

The loss of counts at high count rates is said to be caused by "pulse pile-up." This effect is produced by two or more γ-ray interactions occurring within the resolving time of the scintillation crystal. Pulse pile-up can also cause image distortion if the energy deposited during Compton interactions occurring in different locations is summed to yield an energy equivalent to that of total absorption of a primary γ ray. In this case the events will be registered as a single primary γ-ray interaction at a location between the sites of the Compton interactions. The net effect of this positional distortion is a loss of contrast in the image.

When scintillation cameras are used for dynamic function studies, the loss of counts at high count rates may be a significant problem. In studies of the heart, for example, a bolus of activity on the order of 500 MBq or more may be present in the camera's field of view. This high activity may yield as many as 80,000 scintillations per second in the detector, and possibly even more if a converging collimator is used. At count rates this high, significant count rate losses occur, and quantitative interpretation of count rate data can be misleading.

For purposes of resolving time considerations, a scintillation camera may be considered a two-component system, with a paralyzable component followed by a nonparalyzable component.[4] In a paralyzable counting system, the minimum time required for the independent detection of two separate interactions is extended if an interaction occurs within the recovery interval. With such a system, the count rate may decrease dramatically (i.e., the system may be paralyzed) for samples of exceptionally high activity (Figure 9-6). In a nonparalyzable system, a certain minimum time is required to process each interaction, but this time is not extended by an interaction occurring within the processing interval. With a nonparalyzable system and high sample activity, the count rate may reach a maximum plateau but will not decrease as the activity is increased. Many scintillation cameras exhibit a behavior intermediate between paralyzable and nonparalyzable. In general, the paralyzable component is the part of the camera that precedes the pulse height analyzer and reflects primarily the fluorescence decay time of the scintillation crystal. The nonparalyzable component is the part of the camera, including scalers and data processing equipment, that follows the pulse height analyzer.

A typical NaI(Tl) crystal exhibits a fluorescence decay time of 0.25 μsec; 98% of all light is released within 0.8 μsec.

Several methods have been proposed to correct scintillation camera data for coincidence count rate loss. If the resolving time is known for a particular camera, for example, true counts can be estimated from observed counts, provided that the resolving time is independent of factors such as activity position across the detector face and that the summation of small coincident pulses to produce pulses transmitted by the PHA can be estimated. One widely used method to correct for coincidence count rate loss utilizes the count rate from a small marker source of known activity positioned near the periphery of the camera's field of view. In place of the radioactive marker source, an electronic marker such as a pulse generator may be used.

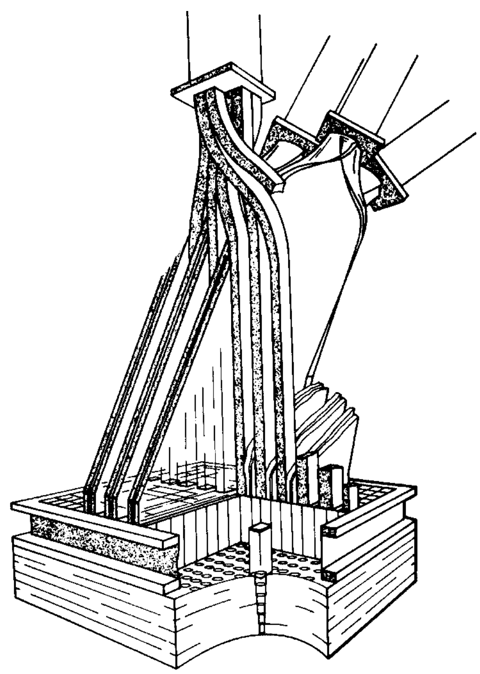
MULTIPLE-CRYSTAL SCINTILLATION CAMERA

The multiple-crystal approach to scintillation camera design was pioneered by Bender and Blau, who introduced the concept of the autofluoroscope in 1963.[5] In this device, 294 discrete NaI(Tl) crystals are arranged in a matrix of 21 columns and 14 rows, with each crystal 0.8 cm by 0.8 cm in cross section and about 4 cm thick.

Later versions of the autofluoroscope were marketed as "System 70."

In a single-crystal scintillation camera, the detection of γ-ray interactions includes determination of the spatial coordinates of the interaction. In a multiple-crystal camera, the detection of an interaction does not require the simultaneous collection of position information, and much higher count rates can be processed without significant count rate loss or positioning errors. In the multiple-crystal camera, the resolving time is determined essentially by the speed with which events can be processed electronically rather than by the fluorescence decay time of the crystal. Most multiple-crystal cameras in clinical use have been reserved for studies such as cardiac first-pass studies where the viewing area is limited and the count rates are very high.

The resolution of the multiple-crystal camera is limited intrinsically by the size of each detector element. However, the resolution can be improved markedly by moving the patient incrementally in the x and y directions so that each resolution element is divided into 16 subelements. With this motion of the scanning bed supporting the patient, the 294 resolution elements are subdivided essentially into $294 \times 16 = 4704$ subelements, with the final image reassembled by computer. For anatomic regions larger than the 15-cm $\times$ 23-cm viewing area of the detector, two successive adjacent 16-position images can be reassembled by computer to furnish a single image containing 9408 picture elements.



MARGIN FIGURE 12-9
Cutaway view of the detector assembly and light pipes for the multicrystal scintillation camera.

SOLID-STATE CAMERA

Nuclear cameras that employ direct digital detectors have been in development for several years. One digital camera introduced commercially just recently employs 4096 3-mm $\times$ 3-mm cadmium–zinc–tellerium [CdZnTe] detector elements. This detector maps γ-ray interactions in the elements directly into an image, so no PM tubes or x- and y-positioning circuits are needed. The detector is 21.6×21.6 cm square, so the field of view is rather limited. For small organs, this version of the direct digital camera provides excellent spatial and contrast resolution.

The commercial version of the direct digital camera described in the text is referred to as Digital Corporation's 2020tc Imager.

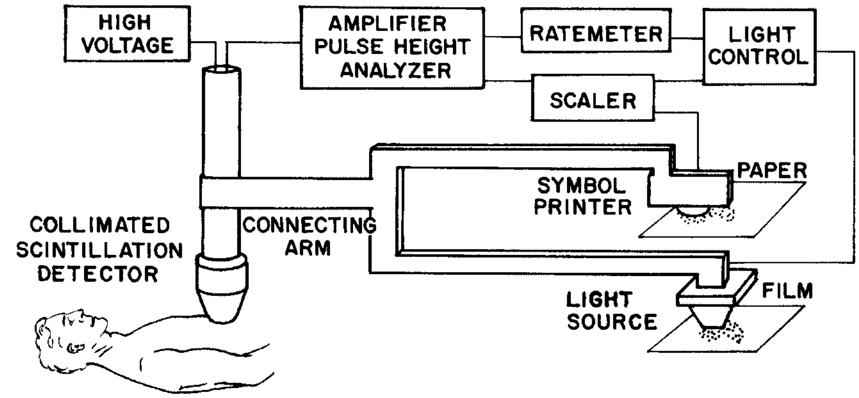


FIGURE 12-8
Rectilinear scanner with mechanical linkage.

RECTILINEAR SCANNER

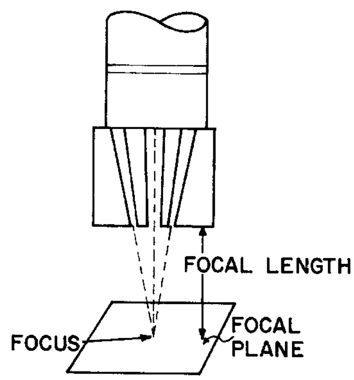
A rectilinear scanner consists of a radiation detector that moves in a plane above or below the patient, and a recording device that moves in synchrony with the detector (Figure 12-8). In earlier scanners, the motion of the detector was synchronized with the recording device by a mechanical linkage. In later versions, electronic rather than mechanical linkage has been used, and the recording device is usually physically separated from the detector.

The first practical rectilinear scanner was developed in 1951 by Cassen and colleagues.[6] In this scanner, a small crystal of calcium tungstate was used as the radiation detector. However, the superior properties of sodium iodide (NaI) for detecting γ rays were quickly recognized, and essentially all scanners now in use employ this type of detector. In earlier scanners, the NaI crystal was typically 7.5 cm in diameter and 5 cm thick. In more recent scanners, the crystal diameter has increased to 13 or 20 cm to improve the efficiency of γ-ray detection. With a larger crystal, images may be obtained more quickly; or alternately, less "noisy" images may be obtained in comparable imaging times. Although the larger crystals collect more γ rays because of their larger field of view, they exhibit a more limited depth of focus, and radioactivity above and below the focal plane of the collimator is imaged less sharply.

Cassen's first rectilinear scanner employed a moving $CaWO_4$ crystal coupled to a PM tube and equipped with a single-hole lead collimator. His first studies were ^{131}I scans of the thyroid.

As the detector moves across the patient, it samples activity from a small region of the patient. This region is defined by a focused collimator and is referred to as the focal plane. The distance between the focal plane and the collimator face is the focal length. Pulse height analysis is used to record the signal. An energy "window" is set to accept only those pulses that correspond to total absorption of the γ rays of interest. Lower-energy pulses correspond to events such as interactions in the crystal of photons that were scattered originally in the patient. Such events would reduce contrast in the final image and are rejected by the PHA.

Advances in stationary scintillation cameras, coupled with the excessive time required to complete a rectilinear scan, have greatly reduced the value and use of rectilinear scanners in recent years.



MARGIN FIGURE 12-10
Multihole focused collimator illustrating the focal length, focal plane, and focus of the collimator.

EMISSION COMPUTED TOMOGRAPHY

Tomography is the process of collecting data about an object from multiple views (projections), and using these data to construct an image of a "slice" through the object. In traditional x-ray tomography, multiple views of an object are built up on a single image receptor in such a way that uninteresting anatomy is blurred while a selected plan or region of the patient is presented as a sharp image. In computed tomography (CT), a computer is used to reconstruct an image of the patient from multiple views. CT using x rays transmitted through the patient is termed *transmission CT*

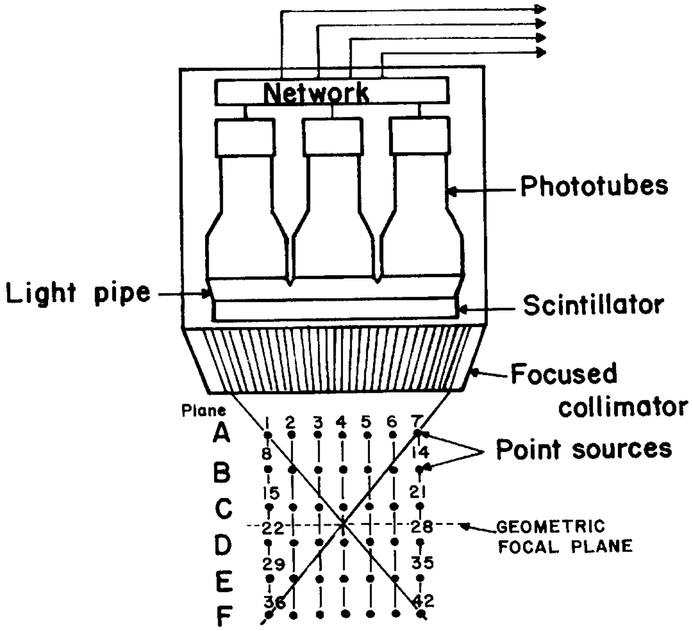


FIGURE 12-9
Detector assembly for multiplane tomographic scanner (From Anger, H.[7] Used with permission.)

(x-ray CT). A similar type of study can be performed in nuclear medicine by detecting photons emitted from a radiopharmaceutical distributed within the body. The term *emission computed tomography* (ECT) refers to this procedure. Production of tomographic images by detection of annihilation photons released during positron decay is referred to as *positron emission tomography* (PET). Tomographic images computed from the registration of interactions of individual γ rays in a crystal is referred to as *single-photon emission computed tomography* (SPECT).

Single-Photon Emission Computed Tomography
Longitudinal-Section SPECT

Also known as focal-plane SPECT, this general category of techniques involves the use of multiple views, and possibly motion, to provide a coronal, sagittal, or oblique slice of information within the patient. A diagram of a longitudinal-section SPECT unit is shown in Figure 12-9. In this unit, two or more views of the object are obtained from different angles. Super-position of data from the multiple views leads to multiple images and therefore blurring of all objects except those at the plane defined by the junction of the viewing areas of the detector. Images may be compiled by simple addition of counts from the multiple views. With the aid of computers, several ingenious techniques are available to reconstruct images from any plane or arbitrary surface within the object. If the fixed spacings between views and the angle between the views at the imaging device are known, the origin of each detected "event" within the crystal may be matched to the "event" seen by another view. In this manner, a source of radiation can be located in three dimensions within the patient.

The first longitudinal-section SPECT unit was built by H. Anger in 1965.[7]

Various methods are used to obtain multiple views for focal-plane SPECT. Special collimators have been designed to acquire the multiple views, and dedicated software has been developed to process the information. Examples of these systems include multiple pinhole techniques[8] sometimes combined with collimator or detector motion.[9–10] Although longitudinal-section SPECT systems continue to have research applications, they have been replaced in clinical nuclear medicine by transverse-section SPECT systems.

Single-Photon Emission Computed Tomography

Transverse-Section SPECT

Development of SPECT can be traced to the work of Kuhl and Edwards in the early 1960s.[11]

This nuclear imaging technique has several features in common with x-ray CT. Multiple views are obtained to compile an image of one or more transverse slices through the patient. In fourth-generation x-ray CT, the x-ray tube rotates and the detectors are stationary. Most transverse-section SPECT systems employ a rotating detector (a scintillation camera) and a stationary source of γ rays (the patient). One, two, or three scintillation cameras are mounted on a rotating gantry. Each scintillation "head" encompasses enough anatomy in the axial direction to permit acquisition of multiple slices in a single scan.

In earlier SPECT units, the detectors rotated in a circular orbit around the body. In newer units, the detectors follow an elliptical path so that they remain close to the skin surface during the entire scan.

One major difference between SPECT and x-ray CT is that a SPECT image represents the distribution of radioactivity across a slice through the patient, whereas an x-ray CT image reflects the attenuation of x rays across a tissue slice. In SPECT, photon attenuation interferes with the imaging process because fewer photons are recorded from voxels at greater depths in the patient. A correction must be made for this property. Methods for attenuation correction usually involve estimating the body contour, sampling the patient from both sides (i.e., 360° rather than 180° rotation), and constructing a correction matrix to adjust scan data for attenuation.

Transverse-section SPECT units can also provide coronal and sagittal images by interpolation of data across multiple transverse planes.

Transverse-section SPECT is useful in several clinical applications including cardiac imaging,[12,13] liver/spleen studies,[14,15] and chest–thorax procedures.[16] It is also important in the imaging of receptor-specific pharmaceuticals such as monoclonal antibodies, because it provides improved detection and localization of small sites of radioactivity along the axial dimension.[17]

For 360° SPECT imaging, a single-head unit must rotate 360°; a dual-head unit must rotate 180°; and a triple-head unit must rotate 120°. Hence, data acquisition is three times faster with a triple-head unit than with a single-head unit.

Quality control for SPECT systems is similar to that for stationary scintillation cameras. In general, uniformity requirements are rather stringent for SPECT, and variations in count rate sensitivity over the image should not exceed 2%. The uniformity of count rate sensitivity should be verified in several orientations of each detector because distortions may be introduced by the influence of the earth's magnetic field on the PM tubes. Also, coupling of the PM tubes to the crystal may change as a detector changes position. Certain parameters are critical, such as assignment of the center of rotation for transverse-section SPECT. If a source of radioactivity is positioned incorrectly in the image matrix for different views during transverse SPECT, it will appear as a ring in the final image. To verify the center of rotation, a point source of activity is placed at the center of rotation and imaged from multiple views. In each view, the source will appear in an individual pixel. The pixel in which the source appears should be the same in views taken from different angles. Another approach to verifying alignment of the center of rotation is to position a point source slightly away from the center of rotation. The point source should appear as a symmetrical donut in the image.

Reconstruction of a SPECT image is performed with the same class of algorithms used in x-ray CT. The most common algorithm is convolution (filtered) backprojection. However, a shift is occurring from convolution algorithms to algebraic iterative approaches for image reconstruction. Iterative algorithms allow easy incorporation of noise-reduction techniques and *a priori* information to improve spatial and temporal information.

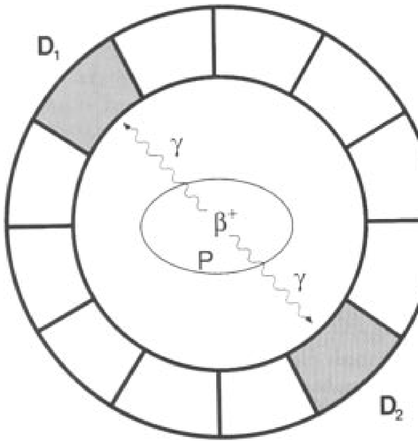
Quality control is critical in transverse-section SPECT. Additional information about SPECT quality-control procedures is available in texts on nuclear imaging.[18–20]

Positron Emission Tomography

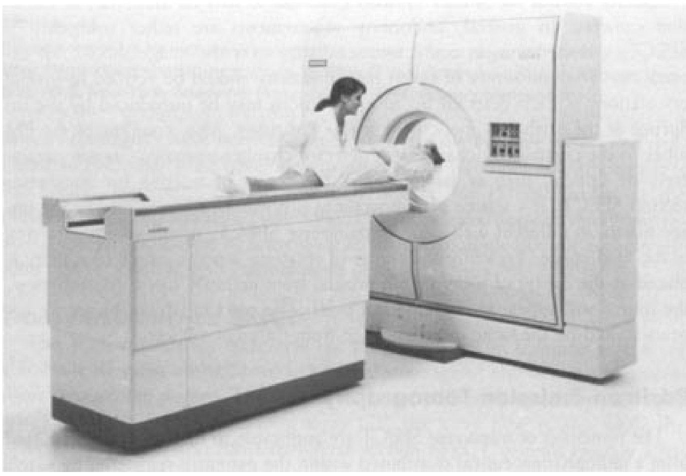
The principle of transverse SPECT is applicable to positron emission tomography (PET). This principle is that radiation emitted from a radiopharmaceutical distributed in the patient is registered by external detectors positioned at different orientations. Then, the distribution of radioactivity is estimated by backprojection methods. In PET, however, the radiation detected is annihilation radiation released as positrons interact with electrons. The directionality of the annihilation photons (two 511-keV annihilation photons emitted in opposite directions) provides a mechanism for localizing the origin of the photons and hence the radioactive decay process that resulted in their emission.

In transverse SPECT, data acquisition can either be continuous [i.e., the detector(s) move continuously] or discontinuous [i.e., the detector(s) stop rotating during data acquisition]. The latter method is referred to as "step and shoot."

In a typical PET system (Figure 12-10) the patient is surrounded by a ring of detectors. When detectors on opposite sides of the patient register annihilation photons simultaneously (e.g., within 10^{-9} seconds), the positron decay process that created the photons is assumed to have occurred along a line between the detectors.



(a)



(b)

FIGURE 12-10
A: In positron emission tomography (PET), two detectors D_1 and D_2 record the interaction of 0.511-MeV photons in coincidence, indicating that a positron decay event has taken place somewhere along a line connecting the two detectors within the patient P. **B:** PET scanner (Siemens ECAT953B brain scanner) at the MRC cyclotron Unit, London. (Part B, courtesy of Siemens Medical Systems, Inc.)

Because the directionality of annihilation photons provides information about the origin of the photons, collimators are not needed in PET imaging. This approach, referred to as *electronic collimation*, greatly increases the detection efficiency of PET imaging.

The PET scanner was developed in the early 1950s by Brownell, Sweet, and Aronow at Massachusetts General Hospital.[21,22]

The PET image reveals the number of decays (counts) occurring in each of the voxels represented by pixels in the image.

Radionuclides suitable for PET are the short-lived (half-lives on the order of seconds, minutes, or a few days) positron-emitting nuclides ^{11}C, ^{13}N, ^{15}O, ^{18}F, ^{62}Cu, ^{68}Ga, and ^{82}Rb. To date, all clinical applications of PET have employed ^{18}F-labeled fluorodeoxyglucose [^{18}FDG]. The positron-emitting isotopes ^{11}C, ^{13}N, and ^{15}O have significant physiologic potential because they can replace atoms in molecules that are essential for metabolism. Thus direct studies of metabolism in tissue are possible (Figure 12-11).

The first multicrystal PET unit was developed in the mid-1970s by TerPogossian, Phelps, and Hoffman[23] of Washington University in St. Louis.

Replacement of an atom in a molecule with its radioactive counterpart is known as *isotopic labeling*.

Positron emitters are produced in a cyclotron. For very short-lived nuclides such as ^{11}C, ^{13}N, and ^{15}O, the cyclotron must be located near the scanner so the nuclides can be used before they decay. Thus the facilities and support personnel required for PET imaging with these radionuclides can be a major financial investment. ^{18}F has a half-life of 1.7 hours and can often be obtained from a nearby supplier. PET imaging confined to use of ^{18}F is less costly but somewhat limited in its universality. There is great interest in further development of PET. The availability of positron sources such as ^{68}Ga that are produced in radionuclide generators, along with advances in miniaturization and simplification of cyclotrons, decreases site planning and facilities

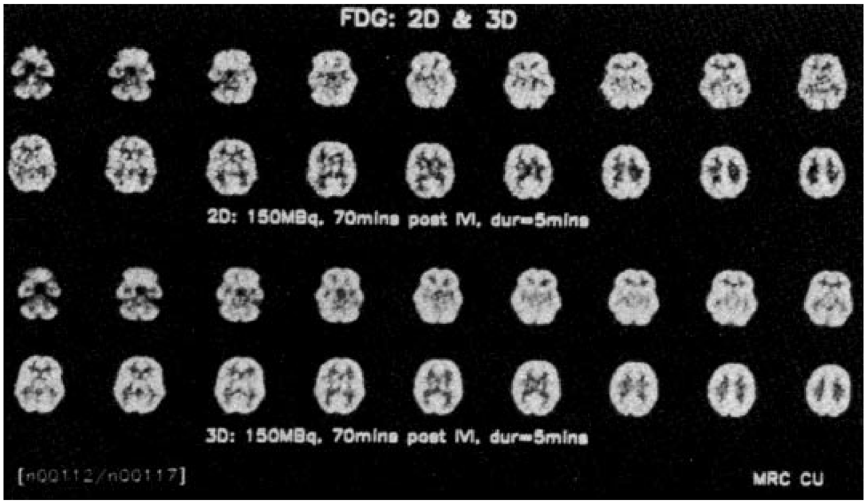


FIGURE 12-11
PET images from a fluorodeoxyglucose study of normal brain glucose metabolism. Signal-to-noise comparison of 2D (with collimators) versus 3D (without collimators) coincidence plane collection of annihilation events. Courtesy of Siemens Medical Systems, Inc.

Self-shielded, computer-controlled, miniaturized cyclotrons with automated chemical synthesis are providing ^{18}F-deoxyglucose (CT-DG)-labeled pharmaceuticals for a variety of PET applications.

PET Radionuclides

Nuclide	$T_{1/2}$ (minutes)
^{11}C	20.4
^{13}N	10
^{15}O	2
^{18}F	110
^{62}Cu	9.8
^{68}Ga	68
^{82}Rb	1.25

support requirements and is stimulating the development of more clinical sites for PET imaging.

It is more challenging to obtain multiple image slices in PET than in SPECT. Most PET scanners employ several detector rings for this purpose. However, attenuation corrections are inherently more accurate in PET than in SPECT because the pair of annihilation photons defines a line through the patient such that the total path length through the patient is always traversed when a decay is detected.

Detector sensitivity is a major design consideration in PET. Bismuth germanate (BGO) crystals are used in place of the NaI(Tl) detectors employed in SPECT, because BGO has a greater intrinsic efficiency for the higher-energy (511 keV) photons. Resolution of PET systems is limited to 5 mm or so by the number of detectors used, distance traveled by the positron from its origin before annihilation, and deviation from exact 180°-annihiliation photon emission. Although PET resolution is poorer than that of x-ray CT or magnetic resonance imaging (MRI), interest in the development of PET is substantial because the physiologic and metabolic information available in a PET study is unique.

A relatively recent development is the integration of PET and SPECT into a single imaging unit. With this integrated unit, SPECT and PET imaging can be performed on the same patient. Also, the merger of PET and x-ray CT into the same imaging unit provides PET images fused with CT images. This approach is particularly useful for tumor identification and localization in oncology, which at this time constitutes the major clinical application of PET. In addition, x-ray CT furnishes a rapid and accurate method of correcting PET images for photon attenuation.

PROBLEMS

*12-1. The uptake of ^{131}I was measured for a patient treated for hyperthyroidism with the same radioactive nuclide 2 months previously. The count rates were as follows:

For a ^{131}I standard on day 0	13,260 cpm
For a ^{131}I standard 24 hour later	12,150 cpm
For the patient on day 0 before ^{131}I administered	6,140 cpm
For the patient 24 hours later	12,840 cpm

Compute the percent uptake of ^{131}I in the thyroid gland. Ignore the contribution of background radiation to the count rates, but include the contribution from residual ^{131}I.

*12-2. A patient was given 10×10^4 Bq of ^{131}I-RISA. A second 10×10^4 Bq sample was diluted to 3000 ml. A 5-ml aliquot of this solution provided a count rate of 3050 cpm. A 5-ml sample of the patient's blood provided a count rate of 1660 cpm. What is the patient's blood volume?

12-3. Describe a method with a radioactive nuclide to determine the volume of a large container.

*12-4. The following count rates were recorded 24 hours after administration of 0.4 MBq of ^{131}I to the patient:

For the patient's thyroid	1831 cpm
For the patient's thyroid shielded by a lead block	246 cpm
For a standard source in a thyroid phantom	3942 cpm
For a standard shielded by a lead block	214 cpm

What is the percent uptake?

*12-5. Line-source response functions were measured for a NaI(Tl) scintillation detector and a multihole focused collimator. Which functions resemble a normal probability curve more closely:
a. A response function measured with a ^{198}Au source (γ ray of 412 keV) or a response function measured with a ^{241}Am source (γ ray of 60 keV)?
b. A response function measured with a ^{51}Cr source (γ ray of 320 keV) or a response function measured with a ^{197}Hg source (γ and x rays of about 70 keV)?

12-6. Explain the use with a scintillation camera of (a) a pinhole collimator, (b) a parallel multihole collimator, (c) a diverging collimator, and (d) a converging collimator. Explain why the sensitivity does not change greatly as the distance varies between the patient and the face of a parallel multihole collimator.

12-7. Explain why the minimum resolution distance of a multiple-crystal camera is about equal to the width of a crystal in the detector mosaic.

12-8. Describe the reasons for limiting the useful range of a scintillation camera from about 70 keV to about 600 keV.

12-9. Explain why, when compared with a parallel multihole collimator, a converging collimator yields superior images for small distributions of radioactivity.

*For those problems marked with an asterisk, answers are provided on p. 492.

SUMMARY

- Radioactive nuclides are widely used to measure rate and volumes in clinical medicine. Examples include:
  - Thyroid uptakes
  - Blood volumes
  - Kidney flow studies
- The single-crystal scintillation camera, together with generator-produced ^{99m}Tc, are the principal factors responsible for the widespread use of nuclear medicine procedures in medicine.
- Collimators provide a one-to-one relationship between the origin of a γ ray in the patient and the site of interaction of the γ ray in the crystal. Types of collimators include:
  - Parallel multihole
  - Pinhole
  - Converging
  - Diverging
- Basic components of a single-crystal scintillation camera include:
  - Detector (crystal, PM tube, preamplifier)
  - Position-encoding matrix
  - Summation and deflection amplifiers
  - Pulse summation circuit
  - Cathode-ray display
- The resolving time of a scintillation camera is the shortest time between individually measurable events in the camera.
  - Emission computed tomography includes:
  - Longitudinal-section SPECT
  - Transverse-section SPECT
  - Positron emission tomography (PET)

REFERENCES

1. Saha, G. *Physics and Radiobiology of Nuclear Medicine, 2nd Edition*. New York, Springer, 2001.
2. Anger, H. Scintillation camera. *Rev. Sci. Instrum.* 1958; **29**:27.
3. Farmelant, M., DeMeester, G., Wilson, D., et al. *J. Nucl. Med.* 1975; **16**:183.
4. Sorensen, J. Deadtime characteristics of Anger cameras. *J Nucl. Med.* 1975; **16**:284.

5. Bender, M., and Blau, M. The autofluoroscope. *Nucleonics* 1963; **21:**52.
6. Cassen, B., et al. Instrumentation for ^{131}I use in medical studies. *Nucleonics* 1951; **9:**46.
7. Anger, H. Tomography and other depth-discrimination techniques, in Hine, G., and Sorenson, J. (eds.), *Instrumentation in Nuclear Medicine*, Vol 2. New York, Academic Press, 1974.
8. Budinger, T. Physical attributes of single-photon tomography. *J. Nucl. Med.* 1980; **21:**579–592.
9. Muehllehner, G. A tomographic scintillation camera. *Phys. Med. Biol.* 1971; **16:**87.
10. Anger, H. *Tomographic Gamma-Ray Scanner with Simultaneous Readout of Several Planes.* UCRL Report 16899. Berkeley, University of California, Press, 1966.
11. Kuhl, D., and Edwards, R. Image separation radioisotope scanning. *Radiology* 1963; **80:**653–662.
12. Garcia, E., et al. Quantification of rotational thallium-201 myocardial tomography. *J. Nucl. Med.* 1985; **26:**17–26.
13. Depasquale, E., et al. Quantitative rotational thallium-201 tomography for identifying and localizing coronary artery disease. *Circulation* 1988; **77:**316–327.
14. Van Heertum, R. Current advances in hepatic SPECT imaging, in *Clinical SPECT Symposium*. American College of Nuclear Physicians, Washington, D.C., 1986, pp. 58–64.
15. Friman, L., and Soderberg, B. Spleen–liver ratio in RES scintigraphy: A comparison between posterior registration and emission computed tomography. *Acta Radiol.* 1987; **28:**439–441.
16. Khan, B., Ell, P., Jarritti, P., et al. Radionuclide section scanning of the lungs in pulmonary embolism. *Br. J. Radiol.* 1981; **54:**586–591.
17. Chatal, J., et al. Comparison of SPECT imaging using monoclonal antibodies with computed tomography and ultrasonography for detection of recurrences of colorectal carcinoma: A prospective clinical study. *J. Nucl. Med.* 1985; **26:**15.
18. Peters, A., and Myers, M. *Physiological Measurements with Radionuclides in Clinical Practice*. New York, Oxford Univ. Press, 1998.
19. Chandra, R. *Nuclear Medicine Physics: The Basics,* 5th edition. Baltimore, Williams & Wilkins, 1998.
20. Powsner, R., and Powsner, E. *Essentials of Nuclear Medicine Physics.* Malden, MA, Blackwell, 1998.
21. Brownell, G., and Sweet, W. Localization of brain tumors. *Nucleonics* 1953; **11:**40–45.
22. Aronow, S., Brownell, G., Lova, S., and Sweet, W. Statistical analysis of eight years' experience in positron scanning for brain tumor localization. *J. Nucl. Med.* 19562; **3:**198.
23. TerPogossian, M., Phelps, M., and Hoffman, E. A positron-emission transaxial tomograph for nuclear imaging (PETT). *Radiology* 1975; **114:**89–98.

CHAPTER

13

RADIOGRAPHY

OBJECTIVES 218

X-RAY FILM 218

Photographic Process 219
Optical Density and Film Gamma 219
Film Speed 221
Film Reciprocity 221
Inherent Optical Density and Film Fogging 221
Processing X-Ray Film 222
Film Resolution 222

INTENSIFYING SCREENS 223

Composition and Properties 223

RADIOGRAPHIC GRIDS 226

Construction of Grids 226
Types of Grids 226
Describing Radiographic Grids 227
Moving Grids 228
Grid Cutoff 229
Air Gap 229
Moving-Slit Radiography 229

MAGNIFICATION RADIOGRAPHY 229

DIGITAL RADIOGRAPHY 230

Discrete Digital Detectors 230
Storage Phosphor 231
Film Scanning 232
Large-Area Digital Image Receptors for Radiography 232

PROBLEMS 233

SUMMARY 234

REFERENCES 234

Medical Imaging Physics, Fourth Edition, by William R. Hendee and E. Russell Ritenour
ISBN: 0-471-38226-4

OBJECTIVES

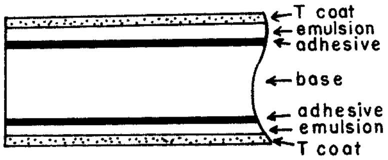
After completing this chapter, the reader should be able to:

- Describe the construction of radiographic film.
- Define the following terms concerning radiographic film:
  - Transmittance
  - Optical density
  - H and D curve
  - Latitude
  - Average gradient
  - Speed
  - Reciprocity
- State the effect of the following upon the H and D curve of film:
  - X-ray tube voltage
  - Development time
  - Development temperature
- State the effect of changes in development time and temperature upon:
  - Average gradient
  - Relative speed
  - Fog
- Describe the principle of operation of the radiographic screen.
- Name five different types of radiographic grids.
- Define the following terms concerning radiographic grids:
  - Grid ratio
  - Contrast improvement factor
  - Selectivity
  - Cutoff
- Explain how the air gap technique may be used to exclude scattered photons from a radiographic image.
- Discuss direct and indirect energy conversion in large-area digital detectors.

The expression "radiography" refers to procedures for recording, displaying, and using information carried to a film by an x-ray beam. Satisfactory images are recorded only if enough information is transmitted to the film by x-rays emerging from the patient. Also, unnecessary information and background "noise" must not interfere with extraction of the information desired. Procedures for recording the information desired and for reducing extraneous information and noise are described in this chapter.

X-RAY FILM

X-ray film is available with an emulsion on one side (single-emulsion film) or both sides (double-emulsion film) of a transparent film base that is about 0.2 mm thick (Margin Figure 13-1). The base is either cellulose acetate or a polyester resin. Single-emulsion film is less sensitive to radiation and is used primarily when exceptionally fine detail is required in the image (e.g., laser printing and mammography). The emulsion is composed of silver halide granules, usually silver bromide, that are suspended in a gelatin matrix. The emulsion is covered with a protective coating (T coat) and is sensitive to visible and ultraviolet light and to ionizing radiation. Film used with intensifying screens is most sensitive to the wavelengths of light emitted by the screens. Nonscreen film is designed for direct exposure to x rays and is less sensitive to visible light. Virtually all film used in diagnostic radiology is designed to be used with intensifying screens.



MARGIN FIGURE 13-1
Cross section of an x-ray film. The base is cellulose acetate or a polyester resin, and the emulsion is usually silver bromide suspended in a gelatin matrix.

Photographic Process

The granules of silver bromide in the emulsion of an x-ray film are affected when the film is exposed to ionizing radiation or to visible light. Electrons are released as energy is absorbed from the incident radiation. These electrons are trapped at "sensitivity centers" in the crystal lattice of the silver bromide granules. The trapped electrons attract and neutralize mobile silver ions (Ag^+) in the lattice. Hence small quantities of metallic silver are deposited in the emulsion, primarily along the surface of the silver bromide granules. Although these changes in the granules are not visible, the deposition of metallic silver across a film exposed to an x-ray beam is a reflection of the information transmitted to the film by the x radiation. This information is captured and stored as a *latent image* in the photographic emulsion.

When the film is placed in a developing solution, additional silver is deposited at the sensitivity centers. Hence the latent image induced by the radiation serves as a catalyst for the deposition of metallic silver on the film base. No silver is deposited along granules that are unaffected during exposure of the film to radiation, and these granules are removed by the sodium thiosulfate or ammonium thiosulfate present in the fixing solution. The fixing solution also contains potassium alum to harden the emulsion and acetic acid to neutralize residual developer present on the film. The degree of blackening of a region of the processed film depends on the amount of free silver deposited in the region and, consequently, on the number of x rays absorbed in the region.

Until the 1960s, the material used for the base of radiographic film was the same material that was used for motion picture film, cellulose nitrate. The same drawback was recognized in both fields; cellulose nitrate turns brown and disintegrates over periods as short as a few years, unless stored under special environmental conditions. This disintegration causes it to emit toxic fumes. Cellulose nitrate radiographic film was the cause of a series of explosions that resulted in number of fatalities at the Cleveland Clinic in May 15, 1929. Today, more stable plastics such as mylar are used for film base material.

Optical Density and Film Gamma

The amount of light transmitted by a region of processed film is described by the *transmittance* T, where

$$T = \frac{\text{Amount light transmitted by a region of film}}{\text{Amount light received at the same location with film removed}} \tag{13-1}$$

The degree of blackening of a region of film is described as the *optical density* (OD) of the region:

$$\text{OD} = \log\left(\frac{1}{T}\right) \tag{13-2}$$

Often, the optical density is referred to as the "density" of the film.

Example 13-1

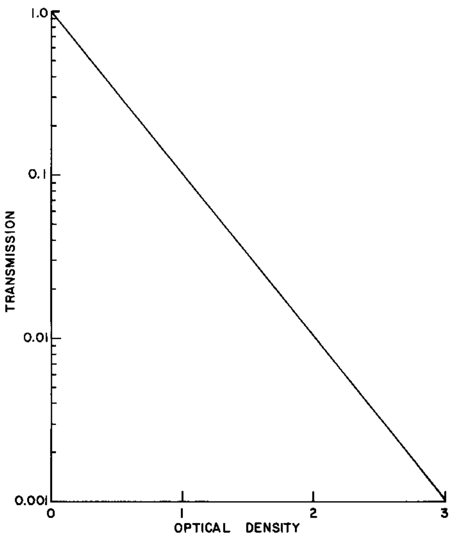
A region of processed film transmits 10% ($T = 0.1$) of the incident light. What is the optical density of the region?

$$\begin{aligned} T &= 0.1 \\ \text{OD} &= \log\left(\frac{1}{T}\right) = \log\left(\frac{1}{0.1}\right) = \log(10) \\ &= 1 \end{aligned}$$

Example 13-2

Two processed films each have optical densities of 1 ($T = 0.1$). What is the resulting optical density if one is placed over the other?

$$T = (0.1)(0.1) = 0.01$$



MARGIN FIGURE 13-2
Relationship between optical density and transmittance of x-ray film.

The optical density is

$$OD = \log\left(\frac{1}{T}\right) = \log\left(\frac{1}{0.01}\right) = \log(100)$$

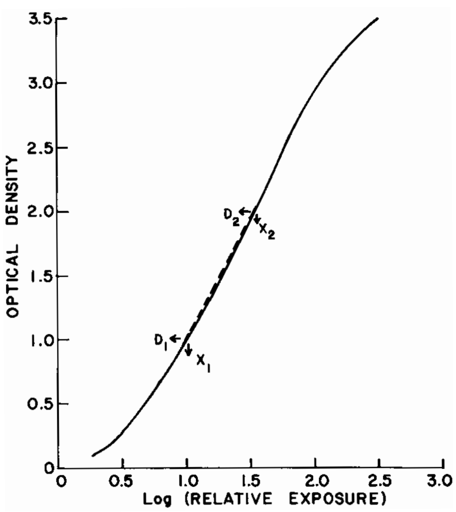
$$OD = 2$$

The relationship between transmittance and optical density is depicted graphically in Margin Figure 13-2. The optical density across most radiographs varies from 0.3 to 2, which corresponds to a range of transmittance from 50% to 1%. Radiographs with an optical density greater than 2 must usually be viewed with a light source more intense than the ordinary viewbox (a "hot light").

The optical density of a particular film or combination of film and intensifying screens may be plotted as a function of the exposure or [log (exposure)] to the film. The resulting curve is termed the *characteristic curve, sensitometric curve*, or *H–D curve* for the particular film or film-screen combination (Margin Figure 13-3). The expression "H–D curve" is derived from the names of Hurter and Driffield, who in 1890 first used characteristic curves to describe the response of photographic film to light. The region below the essentially straight-line portion of the characteristic curve is referred to as the *toe* of the curve. The *shoulder* is the region of the curve above the straight-line portion. A radiograph with optical densities in the region of the toe or the shoulder furnishes an image with inferior contrast (Margin Figure 13-4). The exposure range over which acceptable optical densities are produced is known as the *latitude* of the film. The shape of the characteristic curve for a particular film is affected by the quality of the x-ray beam used for the exposure (Margin Figure 13-5) and by the conditions encountered during development (Margin Figures 13-6 and 13-7). Consequently, the tube voltage used to generate the x-ray beam and the temperature, time, and solutions used for processing the film should be stated when a characteristic curve for a particular film or film–screen combination is displayed.

The radiation exposure to an x-ray film should be sufficient to place the range of optical densities exhibited by the processed film along the essentially straight-line portion of the characteristic curve. The average slope of this portion of the curve is referred to as the *average gradient* of the film:

$$\text{Average gradient} = \frac{D_2 - D_1}{\log X_2 - \log X_1} \qquad \textbf{(13-3)}$$



MARGIN FIGURE 13-3
Characteristic curve for an x-ray film. The average gradient for the film is 1.9 over the density range 1.0 to 2.0.

In Eq. (13-3), D_2 is the optical density resulting from an exposure X_2 and D_1 is the optical density produced by an exposure X_1 (see Margin Figure 13-3). Often, D_1 and D_2 are taken as optical densities of 0.3 and 2.0, respectively. Films with higher average gradients tend to furnish images with higher contrast (more blacks and whites and fewer shades of gray) than do films with lower values of the average gradient. Because contrast and latitude are reciprocally related, films with high average gradients also provide relatively short latitudes and vice versa. The maximum value of the slope of the characteristic curve is termed the *gamma* (γ) of the film. The gamma of a film may be defined explicitly as the slope at the inflection point of the characteristic curve.

Example 13-3

In Margin Figure 13-3, $D_2 = 2.0$, $\log X_2 = 1.52$, $D_1 = 1.0$, and $\log X_1 = 1.00$. What is the average gradient for the film in this region of optical density?

$$\begin{aligned}\text{Average gradient} &= \frac{D_2 - D_1}{\log X_2 - \log X_1}\\ &= \frac{2.0 - 1.0}{1.52 - 1.00}\\ &= 1.9 \qquad \textbf{(13-3)}\end{aligned}$$

Film Speed

Film speed or film sensitivity is a measure of the position of the characteristic curve at a specific value of the exposure axis.

$$\text{Film speed} = \frac{1}{\begin{array}{c}\text{Exposure in R required for an OD}\\ \text{of 1.0 above base density}\end{array}} \tag{13-4}$$

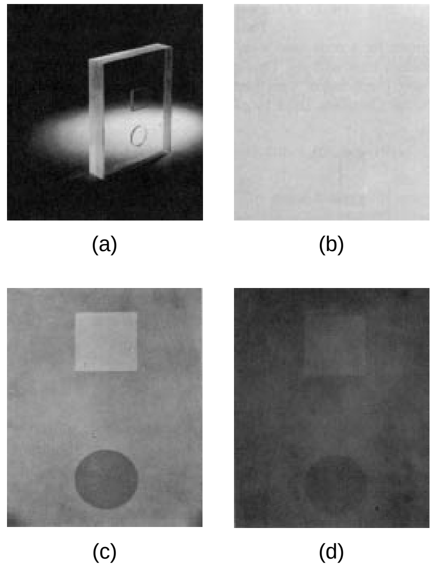
Example 13-4

What is the speed of the x-ray film in Margin Figure 13-3 if 20 mR is required to produce an optical density of 1.0 above base density?

$$\begin{aligned}\text{Film speed} &= \frac{1}{\begin{array}{c}\text{Exposure in R required for an OD}\\ \text{of 1.0 above base density}\end{array}}\\ &= \frac{1}{0.02}\\ &= 50\end{aligned} \tag{13-4}$$

The toe of the H&D curve is short for high-speed film or film-screen combinations and longer for film with lower speed. High-speed film is referred to as "fast" film, and film with lower speed is said to be "slow." The speed of a film depends primarily on the size of the silver bromide granules in the emulsion. A film with large granules is faster than a film with smaller granules. Speed is gained by requiring fewer x-ray or light photons to form an image. Hence, a fast film furnishes a "noisier" image in which fine detail in the object may be less visible (see Chapter 16).

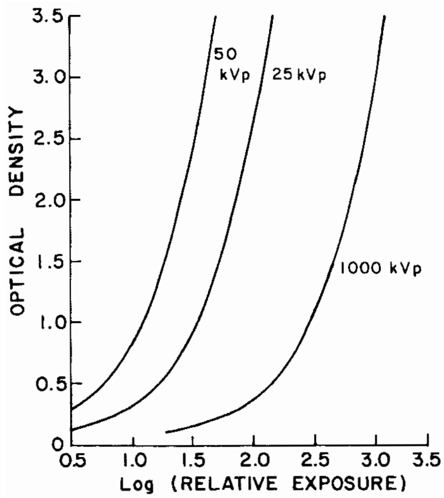
Film speed varies with the energy of the x rays used to expose the film (Margin Figure 13–8). For nonscreen film, the film speed is greatest when the spectral distribution of the x-ray beam is centered over the K-absorption edge (25 keV) of silver.



MARGIN FIGURE 13-4
Photograph of a plastic test object **(A)** and radiographs of the object obtained with screen-type films and an exposure on the toe **(B)**, middle **(C)**, and shoulder **(D)** of the characteristic curve. Contrast of the image is highest for the exposure on the straight-line portion of the curve. (From *Sensitometric Properties of X-Ray Films*. Rochester, NY, Radiography Markets Division, Eastman Kodak Company. Used with permission.)

Film Reciprocity

For most exposure rates encountered in radiography, the shape of the characteristic curve is unaffected by changes in the rate of exposure of the film to x radiation. The independence of optical density and exposure rate is referred to as the *reciprocity law*. The reciprocity law sometimes fails if a film is exposed at either a very high or a very low exposure rate. In these situations, a particular exposure delivered at a very high or very low rate provides an optical density less than that furnished by an exposure delivered at a rate nearer optimum. However, a decrease in optical density is usually not noticeable unless the film is exposed with calcium tungstate intensifying screens and unless the exposure rate differs from the optimum by a factor of at least 8.[1] For film exposed with rare earth intensifying screens, failure of the reciprocity law has been observed over a more restricted range of exposure rates.[2]

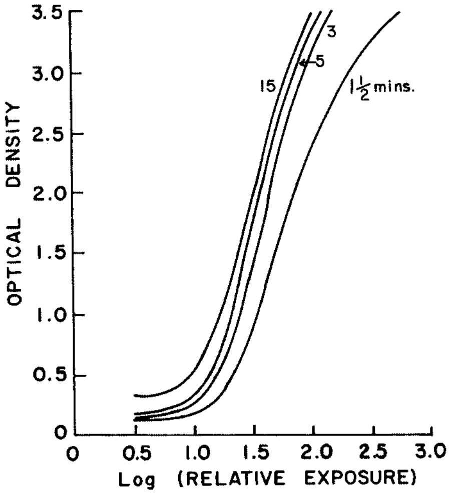


MARGIN FIGURE 13-5
Characteristic curves for a nonscreen x-ray film exposed to heavily filtered x-ray beams generated at different tube voltages. Developing conditions were constant for all curves. (From *Sensitometric Properties of X-Ray Films*. Rochester, NY, Radiography Markets Division, Eastman Kodak Company. Used with permission.)

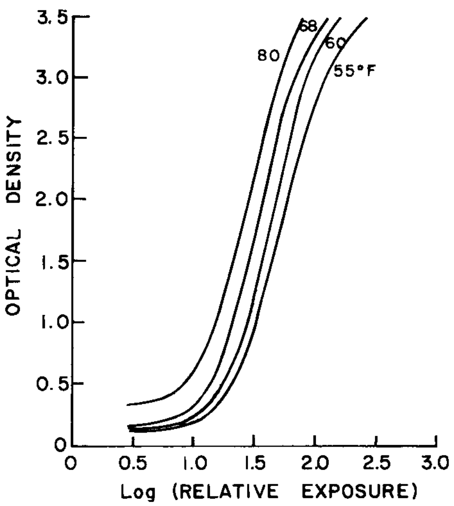
Inherent Optical Density and Film Fogging

The optical density of film processed without exposure to radiation is referred to as the *inherent optical density (base density)* of the film. The inherent optical density is a reflection primarily of the absorption of light in the film base, which may be tinted slightly. Most x-ray film has an inherent optical density of about 0.07.

Fogging of an x-ray film is caused by the deposition of metallic silver without exposure to radiation and by the undesired exposure of a film to radiation (usually background radiation). For example, exposure as small as 1 mR to a high-speed film may produce a significant amount of fogging. The optical density attributable



MARGIN FIGURE 13-6
Characteristic curves for a screen-type x-ray film developed for a series of times at 68°F. (From *Sensitometric Properties of X-Ray Films*. Rochester, NY, Radiography Markets Division, Eastman Kodak Company. Used with permission.)



MARGIN FIGURE 13-7
Characteristic curves for a screen-type x-ray film developed for 5 minutes at a series of developing temperatures. (From *Sensitometric Properties of X-Ray Films*. Rochester, NY, Radiography Markets Division, Eastman Kodak Company. Used with permission.)

to fog may be as low as 0.05 for fresh film but increases with the storage time of the film, particularly if the temperature of the storage facility is not below room temperature. An inherent-plus-fog optical density greater than 0.2 is considered excessive.[3]

Processing X-Ray Film

Properties of x-ray film that affect the quality of a radiographic image include contrast, speed, and the ability of the film to record fine detail. Any two of these properties may be improved at the expense of the third, and a particular type of film is chosen only after a compromise has been reached among the various properties.

The properties of x-ray film are affected to some extent by the conditions encountered during processing. The speed of an x-ray film increases initially with development time. With a long development time, however, film speed reaches a plateau and even may decrease (Margin Figure 13-9). Fogging of an x-ray film increases with time spent in the developing solution. The average gradient of a film increases initially with development time and then decreases. The development time recommended by the supplier of a particular type of x-ray film is selected to provide acceptable film contrast and speed with minimum fogging of the film.

The temperature of the developing solution also affects film speed, fog, and contrast (Margin Figure 13-10). For films processed manually, a temperature of 68°F is usually recommended for the developing solution, and the development time is chosen to furnish satisfactory contrast and speed with minimum fogging. With automatic, high-speed processors, the temperature of the developing solution is elevated greatly (85°F to 95°F), and the development time is reduced. The relationship between development time and temperature of the developing solution is illustrated in Table 13-1.

Contrast, speed, and fog are not affected greatly by changes in the fixation time or the temperature of the fixing solution. However, a decrease in film speed and contrast and an increase in film fog may be noticeable if the fixation time is prolonged greatly.

Film Resolution

The *resolution* (detail, definition, sharpness) of radiographic and fluoroscopic images is discussed in detail in Chapter 16. In general, resolution describes the ability of an x-ray film or a fluoroscopic screen to furnish an image that depicts differences in the transmission of radiation by adjacent small regions of a patient. A detector with "good" or "high" resolution provides an image with good resolution of fine structure in the patient. A detector with "poor" or "coarse" resolution (unsharpness) furnishes an image that depicts only relatively large anatomic structures in the patient.

TABLE 13-1 Relationship Between Development Time and Temperature of Developing Solution for Particular X-Ray Film

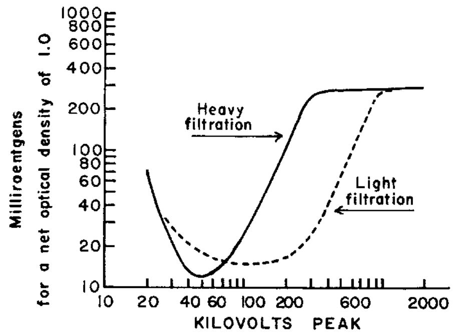
Developer Temperature (°F)	*Development Time (min)*
60	9
65	6
68	5
70	4 1/4
75	3

Source: Seemann, H. *Physical and Photographic Principles of Medical Radiography*. New York, John Wiley & Sons, 1968, p. 69. Used with permission.

Some film-screen manufacturers use their own method of specifying relative speed. A "par" or average system is identified, given a value of 100, and the speeds of other systems are normalized to this reference value. For example, very fast film-screen systems might have relative speeds of 800–1000.

INTENSIFYING SCREENS

For an x-ray beam of diagnostic quality, only about 2% to 6% of the total energy in the beam is absorbed in the emulsion of an x-ray film exposed directly to the beam. The amount of energy absorbed is even smaller for x rays of greater energy. Consequently, direct exposure of film to x rays is a very inefficient utilization of energy available in the x-ray beam. This procedure is used only when images with very fine detail are required. For most radiographic examinations, the x-ray film is sandwiched between *intensifying screens*. The intensifying screens furnish a light image that reflects the variation in exposure across the x-ray beam. This light image is recorded by film that is sensitive to the wavelengths of light emitted by the screen. The mechanism for the radiation-induced fluorescence of an x-ray intensifying screen is similar to that discussed in Chapter 12 for an NaI(T1) scintillation crystal.



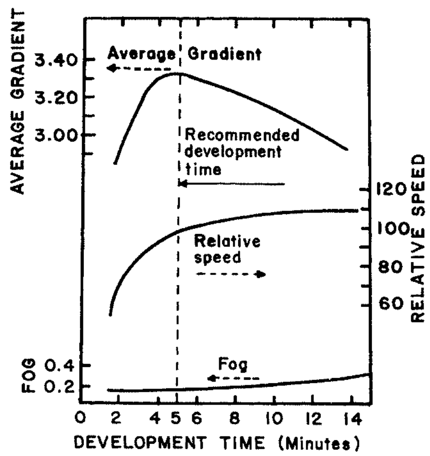
MARGIN FIGURE 13-8
Exposure for an optical density of 1.0 above base density, expressed as a function of the quality of the x-ray beam used to expose the film. The speed of the film is the reciprocal of the exposure in roentgens required for a density of 1.0 above the base density. (From *Sensitometric Properties of X-Ray Films*. Rochester, NY, Radiography Markets Division, Eastman Kodak Company. Used with permission.)

Composition and Properties

The construction of an intensifying screen is diagramed in Margin Figure 13-11. The backing is cardboard, plastic, or, less frequently, metal. The backing is coated with a white pigment that reflects light. The reflecting layer is covered by an active layer that is composed of small granules of fluorescent material embedded in a plastic matrix. The granule diameters range from about 4 to about 8 μm. The thickness of the active layer ranges from perhaps 50 μm for a detail screen to about 300 μm for a fast screen. The active layer is protected by a coating about 0.001 in. thick that is transparent to the light produced in the active layer.

Desirable properties of an x-ray intensifying screen include the following:

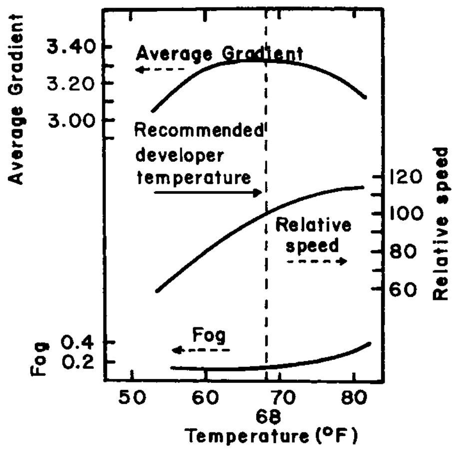
1. A high attenuation coefficient for x rays of diagnostic quality.
2. A high efficiency for the conversion of energy in the x-ray beam to light.
3. A high transparency to light released by the fluorescent granules.
4. A low refractive index so that light from the granules will be released from the screen and will not be reflected internally.
5. An insensitivity to handling.
6. An emission spectrum for the radiation-induced fluorescence that matches the spectral sensitivity of the film used.
7. A reasonably short time for fluorescence decay.
8. A minimum loss of light by lateral diffusion through the fluorescent layer. To reduce this loss of light, the fluorescent layer is composed of granules and is not constructed as a single sheet of fluorescent material.
9. Low cost.



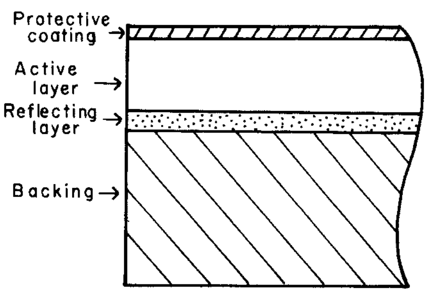
MARGIN FIGURE 13-9
Effect of development time on film speed, film gradient, and fog. (From *Sensitometric Properties of X-Ray Films*. Rochester, NY, Radiography Markets Division, Eastman Kodak Company. Used with permission.)

Crystalline calcium tungstate, with an emission spectrum peaked at 400 nm, was used in most x-ray intensifying screens until the early 1970s. For most calcium tungstate screens, the efficiency is about 20% to 50% for the conversion of energy in the x-ray beam to light.[4,5] Barium lead sulfate, with an emission spectrum peaked in the ultraviolet region (360 nm), is also used as a fluorescent material in intensifying screens. This material is particularly useful for radiography at higher tube voltages.

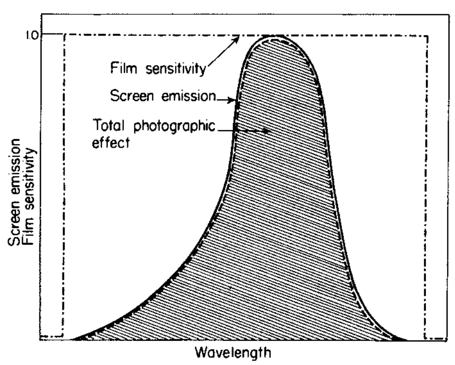
The K-absorption edge (69.5 keV) of tungsten, the principal absorbing element in calcium tungstate screens, is above the energy of most photons in a diagnostic x-ray beam. Increased absorption of diagnostic x rays would occur if tungsten were replaced by an absorbing element of reduced Z with a K-absorption edge in the range of 35 to 50 keV. Elements with K-absorption edges in this energy range include the elements gadolinium, lanthanum, and yttrium. These elements, sometimes referred to as "rare-earth elements" and complexed in oxysulfide or oxybromide crystals embedded in a plastic matrix, have some advantages over calcium tungstate for use in intensifying screens. Rare-earth screens exhibit not only an increased absorption of diagnostic x rays but also an increased efficiency in the conversion of absorbed x-ray energy to



MARGIN FIGURE 13-10
Effect of temperature of the developing solution on film speed, fog, and inherent contrast. (From *Sensitometric Properties of X-Ray Films*. Rochester, NY, Radiography Markets Division, Eastman Kodak Company. Used with permission.)



MARGIN FIGURE 13-11
Construction of a typical intensifying screen.



MARGIN FIGURE 13-12
Total photographic effect obtained by combining an intensifying screen and a photographic film with a wavelength sensitivity that extends above and below the emission spectrum for the screen.

light. Hence rare-earth screens are faster than their calcium tungstate counterparts. This increased speed facilitates one or more of the following changes in radiographic technique:

1. Reduced exposure time and decreased motion unsharpness in the image.
2. Reduced tube current allowing more frequent use of small focal spots.
3. Reduced tube voltage and improved contrast in the image.
4. Reduced production of heat in the x-ray tube.
5. Reduced patient exposure.

Some rare earth screens emit blue light and can be used with conventional x-ray film; others emit yellow-green light and must be used with special yellow-green-sensitive film.[6]

The efficiency with which energy in an x-ray beam is converted to light increases with the thickness of the active (fluorescent) layer. Hence the efficiency of energy conversion is greater for a screen with a thick active layer (i.e., a fast screen) than for a screen of medium speed (a par-speed screen) or for a slow screen (a detail screen). However, the resolution of the radiographic image decreases as the thickness of the active layer increases. For example, for typical calcium tungstate screens the maximum resolution is 6 to 9 lines per millimeter for a fast (thick active layer) screen. Using the same screen material, resolution increases to 8 to 10 lines per millimeter for a par-speed screen (thinner), and 10 to 15 lines per millimeter for a detail screen (thinnest) when measured under laboratory conditions. The resolution obtained clinically with these screens is usually considerably less.

The wavelength of light emitted by an intensifying screen should correspond closely with the spectral sensitivity of the film used. Usually, the sensitivity of the film extends to wavelengths both above and below those emitted by the screen (Margin Figure 13-12). The total photographic effect is decreased if the spectral sensitivity of the film is matched too closely to the emission spectrum for the screen (Margin Figure 13-13).

The *intensification factor* of a particular screen-film combination is the ratio of exposures required to produce an optical density of 1.0 without and with the screen in position.

$$\text{Intensification factor} = \frac{\begin{array}{c}\text{Exposure required to produce}\\ OD\text{ 1.0 without screen}\end{array}}{\begin{array}{c}\text{Exposure required to produce}\\ OD\text{ 1.0 with screen}\end{array}} \qquad \textbf{(13-5)}$$

The intensification factor for a particular screen-film combination varies greatly with the energy of the x-rays used for the radiation exposure, but a range of 50 to 100 is common. Therefore patient exposure is reduced substantially through the use of screens.

Usually, a double-emulsion film is sandwiched in a cassette between two intensifying screens. In some cassettes, the fluorescent layer behind the film is thicker than the layer in front. The resolution of the radiographic image is severely reduced if both screens are not in firm contact with the entire surface of the film.

The major advantage of x-ray intensifying screens is a reduction of the exposure required to form an image of acceptable quality. The radiation exposure of patients is similarly reduced. Because exposure depends linearly upon the product of tube current (milliamperes [mA]) and exposure time (seconds [s]), any reduction of either current or time that leads to a lower product (mA·s) may be used. Shorter exposures (reduced time) may lead to improved resolution by reducing voluntary and involuntary patient motion. Thus, the use of faster screens is generally associated with shorter exposure times.

To summarize factors that describe the speed of fluorescent screens, we may compare conventional calcium tungstate screens with rare-earth screens. Absorption of x-ray photons in calcium tungstate screens varies from 20% to 40% (for par-speed to high-speed screens), while the absorption is approximately 40% to 60% for rare-earth screens. The percentage of absorbed energy converted to light (the conversion efficiency) is approximately 5% for calcium tungstate and 20% for the rare earths. Therefore, rare-earth screens not only absorb more x-ray photons but also convert more of the absorbed energy into visible light.

Example 13-5

Compare the number of visible light photons emitted by a calcium tungstate screen with the number emitted by a gadolinium oxysulfide screen. Assume that 100 x-ray photons, each having an energy of 30 keV, strike the screen. The following data are given:

	Calcium Tungstate	Gadolinium Oxysulfide
Absorption (%)	40	60
Conversion (%)	5	20
Spectral emission peak (nm)	420	550

The energy per visible photon emitted (here we make the simplifying assumption that all photons are emitted at the spectral peak energy) is

$$\text{keV} = \frac{1.24}{\lambda(\text{nm})} = \frac{1.24}{420} = 3 \times 10^{-3}\ \text{keV} = 3\ \text{eV for calcium tungstate}$$

$$= \frac{1.24}{550}$$

$$= 2.25 \times 10^{-3}\ \text{keV} = 2\ \text{eV for gadolinium oxysulfide}$$

The energy converted to visible light photons emitted from each screen is

$$\text{Energy emitted} = (100)(30 \times 10^3\ \text{eV})\,(0.40)\,(0.05)$$

$$= 6 \times 10^4\ \text{eV for calcium tungstate}$$

$$= (100)(30 \times 10^3\ \text{eV})\,(0.60)\,(0.20)$$

$$= 36 \times 10^4\ \text{eV for gadolinium oxysulfide}$$

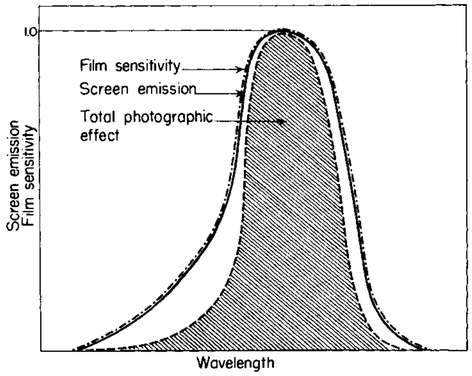
The number of visible photons emitted from each screen is

$$\text{No. of photons emitted}\ \frac{6 \times 10^4\ \text{eV}}{3\ \text{eV/photon}} = 2 \times 10^4\ \text{photons for calcium tungstate}$$

$$\frac{36 \times 10^4\ \text{eV}}{2\ \text{eV/photon}} = 18 \times 10^4\ \text{photons for gadolinium oxysulfide}$$

Thus the lower energy per visible photon and the higher absorption and conversion efficiencies all act to produce more visible photons per incident photon for the gadolinium oxysulfide screen than for the calcium tungstate screen.

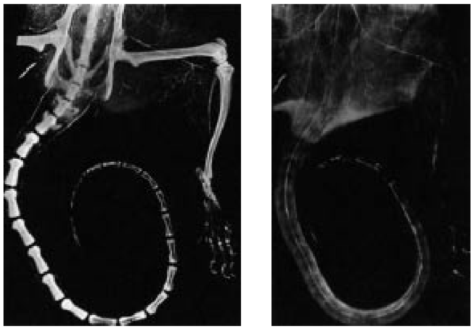
A complete evaluation of screen performance requires analysis of not only speed but also detail and noise (quantum mottle, discussed in Chapter 16). The advantages of higher speed screens (i.e., reduced motion unsharpness, reduced patient dose) may be offset in some clinical situations by the disadvantages of increased noise or loss of detail. Shown in Margin Figure 13-15 are radiographs obtained with and without a fast calcium tungstate screen. The film exposed without a screen required an exposure of 125 mA·s. An exposure of 7 mA·s was required for the film exposed with the intensifying screen.



MARGIN FIGURE 13-13
The total photographic effect is reduced if the spectral sensitivity of the film is matched too closely with the emission spectrum of the intensifying screen.

Industrial radiography is used to determine the structural integrity of steel and other materials. When thicknesses of steel greater than 0.25 inch are radiographed, voltages above 130 kVp are used and the intensifying screen is usually constructed from lead foil. The lead converts the photon energy to kinetic energy of electrons, which go on to deposit energy in the film. For photon energies below 2 MeV, the lead foil need be only 0.004 to 0.006 inch thick.

Neutron Radiography:
Radiographs obtained by exposing objects to slow neutrons (kinetic energy = 0.025 eV) are remarkably dissimilar from those obtained by exposing the same objects to x rays (Margin Figure 13-14). Slow neutrons are attenuated primarily during interactions with low-Z nuclei. In tissue, neutrons are attenuated more rapidly in muscle and fat than in bone because the concentration of low-Z nuclei (primarily hydrogen) is greater in muscle and fat. Neutron radiographs display excellent contrast between air and soft tissue and between soft tissue and structures that contain a contrast agent (e.g., Gd_2O_3) for slow neutrons. The contrast is poor between fat and muscle but may be improved by exposing the specimen to heavy water (water with an increased concentration of deuterium). Bone transmits slow neutrons readily, and neutron radiographs of regions containing bone furnish excellent detail of soft tissue without the distracting image of surrounding bone.



(a) (b)

MARGIN FIGURE 13-14
Arteriograms of a rat leg, paw, and tail obtained by using Gd_2O_3 as a contrast medium. **A:** Radiograph obtained with 20-kVp x-rays. **B:** Slow neutron radiograph exposed with an antiscatter grid. (From Brown, M., Allen, J., and Parks, P. *Biomed. Sci. Instrum.* 1969; **6**. p. 100 & p. 102. Used with permission.)

(a) (b)

MARGIN FIGURE 13-15
Effect of intensifying screens on exposure time and image detail. **A:** Radiograph exposed without a screen and requiring 125 mA·s. **B:** Radiograph exposed with a fast calcium tungstate screen and requiring 7 ma·s.

The first American photographic plate (before screens were used) was manufactured in 1896 by John Carbutt and Arthur W. Goodspeed of Philadelphia. Called the Roentgen x-ray plate, it used a thicker and heavier silver emulsion than was used for ordinary photography and thereby permitted a dramatic reduction in time of exposure. The time required to obtain an image of extremities was reduced from an hour to "only" 20 minutes.

The influence of scattered radiation on the radiographic image is reduced somewhat when intensifying screens are used because scattered x rays with reduced energy are absorbed preferentially in the upper portions of the phosphor layer of the screen. Many of the light photons produced by these scattered x rays are absorbed before they reach the film.

A few precautions are necessary to prevent damage to an intensifying screen. For example, cassettes with intensifying screens should be loaded carefully with film to prevent scratching of the screen surface and accumulation of electrical charge, which can produce images of static discharges on the radiograph. Moisture on the screen may cause the film to adhere to the screen. When the film and screen are separated, part of the screen may be removed. A screen may be stained permanently by liquids such as developing solution, and care must be taken to ensure that liquids are not splashed onto the screen. The surface of the screen must be kept clean and free from lint and other particulate matter. Although screens may be washed with a soap solution or a weak wetting agent, organic solvents should be avoided because the active layer may be softened or otherwise damaged by these materials.

RADIOGRAPHIC GRIDS

Information is transmitted to an x-ray film by unattenuated primary radiation emerging from a patient. Radiation scattered within the patient and impinging on the film tends to conceal this information by producing a general photographic fog on the film. The amount of radiation scattered to a film increases with the volume of tissue exposed to the x-ray beam. Hence a significant reduction in scattered radiation may be achieved by confining the x-ray beam to just the region of interest within the patient. In fact, proper collimation of an x-ray beam is essential to the production of radiographs of highest quality.

Much of the scattered radiation that would reduce the quality of the radiographic image may be removed by a radiographic grid between the patient and the x-ray film or screen-film combination. A radiographic grid is composed of strips of a dense, high-Z material separated by a material that is relatively transparent to x rays. The first radiographic grid was designed by Bucky in 1913. The ability of a grid to remove scattered radiation is depicted in Margin Figure 13-15.

Construction of Grids

Ideally, the thickness of the strips in a radiographic grid should be reduced until images of the strips are not visible in the radiographic image. Additionally, the strips should be completely opaque to scattered radiation and should not release characteristic x-rays ("grid fluorescence") as scattered x-ray photons are absorbed. These requirements are satisfied rather well by lead foil about 0.05 mm thick, and this material is used for the strips in most radiographic grids.

The interspace material between the grid strips may be aluminum, fiber, or plastic. Although fiber and plastic transmit primary photons with almost no attenuation, grids with aluminum interspaces are sturdier. Also, grids with aluminum interspaces may furnish a slightly greater reduction of the scattered radiation because scattered x-ray photons that escape the grid strips may be absorbed in the aluminum interspaces.[7]

Types of Grids

Radiographic grids are available commercially with "parallel" or "focused" grid strips in either linear or crossed grid configurations (Margin Figures 13-17 and 13-18). When a focused grid is positioned at the correct distance from the target of an x-ray tube, lines through the grid strips are directed toward a point or "focus" on the target. The focus-grid distance approaches infinity for a grid with parallel strips. With a parallel grid positioned at a finite distance from an x-ray tube, more primary x rays are attenuated along the edge of the radiograph than at the center. Consequently, the optical density decreases slightly from the center to the edge of a radiograph exposed with a parallel grid. The uniformity of optical density is improved in a radiograph exposed with a focused grid, provided that the grid is positioned correctly.

A linear grid is constructed with all parallel or focused grid strips in line (Margin Figure 13-18). A crossed grid is made by placing one linear grid on top of another, with the strips in one grid perpendicular to those in the other. In most situations, a crossed grid removes more scattered radiation than does a linear grid with the same grid ratio (see below) because a linear grid does not absorb photons scattered parallel to the grid strips. However, a linear grid is easier to use in situations where proper alignment of the x-ray tube, grid, and film cassette is difficult.

Describing Radiographic Grids

A grid usually is described in terms of its *grid ratio* (Margin Figure 13-19):

$$\text{Grid ratio} = \frac{\text{Height of grid strips}}{\text{Distance between grid strips}} \qquad \textbf{(13-6)}$$

$$r = \frac{h}{d}$$

The grid ratio of a crossed grid is $r_1 + r_2$, where r_1 and r_2 are the grid ratios of the linear grids used to form the crossed grid. Radiographic grids are available commercially with grid ratios as high as 16. However, grids with ratios of 8 to 12 are used most frequently because the removal of scattered radiation is increased only slightly with grids with a higher ratio. Also, grids with a high grid ratio are difficult to align and require a greater exposure of the patient to radiation. The effectiveness of a radiographic grid for removing scattered radiation is illustrated in Margin Figure 13-20 as a function of grid ratio.

The improvement in radiographic contrast provided by grids with different grid ratios is depicted in Table 13-2. From data in this table, it is apparent that (1) all grids improve radiographic contrast significantly, (2) the effectiveness of a grid for improving radiographic contrast increases with the grid ratio, (3) the increase in radiographic contrast provided by a grid is less for an x-ray beam generated at a higher voltage, (4) a crossed grid removes scattered radiation more effectively than a linear grid with an equal grid ratio, and (5) the radiation exposure of the patient increases with the grid ratio. The exposure must be increased because the film is exposed to fewer scattered x-ray photons. Also, more primary x-ray photons are absorbed by the grid strips.

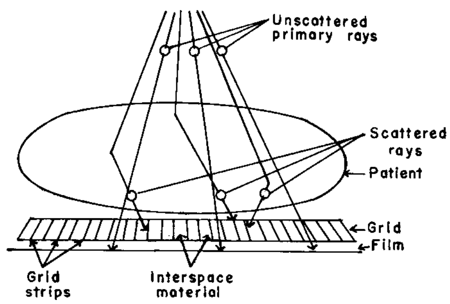
TABLE 13-2 Relative Increase in Radiographic Contrast and Radiation Exposure for Grids with Different Ratios[a]

Type of Grid and Grid Ratio	*Improvement in Contrast*			*Increase in Exposure*		
	70 kVp	*95 kVp*	*120 kVp*	*70 kVp*	*95 kVp*	*120 kVp*
None	1	1	1	1	1	1
5 linear	3.5	2.5	2	3	3	3
8 linear	4.75	3.25	2.5	3.5	3.75	4
12 linear	5.25	3.75	3	4	4.25	5
16 linear	5.75	4	3.25	4.5	5	6
5 cross	5.75	3.5	2.75	4.5	5	5.5
8 cross	6.75	4.25	3.25	5	6	7

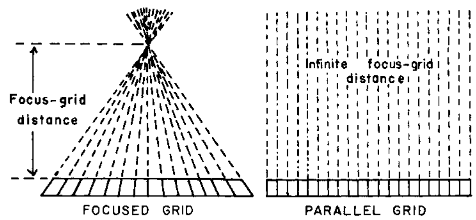
[a] The thickness of the grid strips and interspaces were identical for all grids.

Source: Characteristics and Applications of X-Ray Grids. Cincinnati, Liebel-Flarsheim Co., 1968. Used with permission.

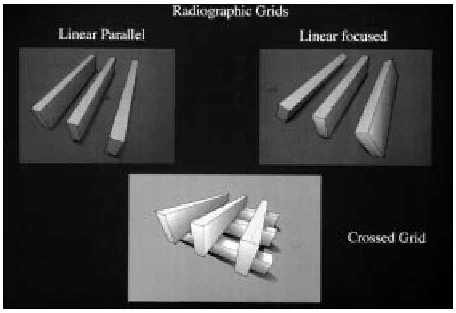
Michael Pupin of Columbia University was the first person to use an intensifying screen in medical radiography. In early 1896 a well-known lawyer had sustained a shotgun blast to the hand, and he was sent to Pupin by a surgeon to obtain a radiograph that would aid in localization and removal of the shotgun pellets. The first two attempts at radiography were unsuccessful because the patient could not endure the hour-long process of exposure. Fortunately, Pupin's friend, Thomas Edison, had recently sent him a calcium tungstate fluoroscopic screen. Pupin placed the fluoroscopic screen upon the photographic plate and obtained a clear image of the pellets and the bones of the hand with an exposure that lasted only a few seconds. The lawyer offered to establish a fellowship for Pupin at a local establishment that would entitle him to two free drinks per day for life. (*Source:* Pupin, M. *From Immigrant to Inventor*. New York, Scribners, 1923.)



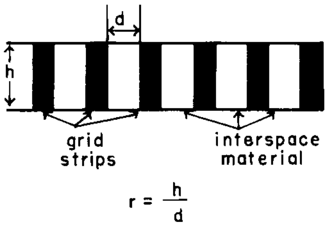
MARGIN FIGURE 13-16
A radiographic grid is used to remove scattered radiation emerging from the patient. Most primary photons are transmitted through the grid without attenuation.



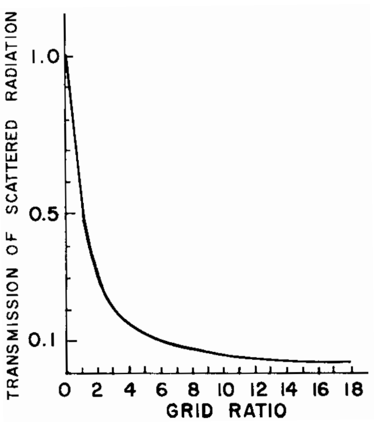
MARGIN FIGURE 13-17
Focused and parallel grids.



MARGIN FIGURE 13-18
Types of focused and parallel grids.



MARGIN FIGURE 13-19
The grid ratio r is the height h of the grid strips divided by the distance between the strips.



MARGIN FIGURE 13-20
The effectiveness of a radiographic grid for removing scattered radiation is pronounced for small grid ratios but increases only gradually with grid ratios above 8. (Courtesy of S. Ledin and the Elema Shönander Co.)

In 1915, G. Bucky developed the moving grid, which, after it was perfected by H. E. Potter in 1919, became known as the Potter–Bucky diaphragm.

A radiographic grid is not described explicitly by the grid ratio alone because the grid ratio may be increased by either increasing the height of the grid strips or reducing the width of the interspaces. Consequently, the number of *grid strips per centimeter* (or per inch) usually is stated with the grid ratio. Grids with many strips ("lines") per inch produce shadows in the radiographic image that may be less distracting than those produced by thicker strips in a grid with fewer strips per inch. Grids with as many as 110 strips per inch are available commercially. The *lead content* of the grid in units of grams per square centimeter sometimes is stated with the grid ratio and the number of strips per inch.

The *contrast improvement factor* for a grid is the quotient of the maximum radiographic contrast obtainable with the grid divided by the maximum contrast obtainable with the grid removed. The contrast improvement factor may be used to compare the effectiveness of different grids for removing scattered radiation. The contrast improvement factor for a particular grid varies with the thickness of the patient and with the cross-sectional area and energy of the x-ray beam. Usually the contrast improvement factor is measured with a water phantom 20 cm thick irradiated by an x-ray beam generated at 100 kVp.[9]

The *selectivity* of a grid is the ratio of primary to scattered radiation transmitted by the grid. The efficiency of a grid for removing scattered radiation is described occasionally as *grid cleanup*. Grids may be described as "heavy" or "light," depending upon their lead content. One popular but rather ambiguous description of grid effectiveness is the *Bucky factor*, defined as the exposure to the film without the grid divided by the exposure to the film with the grid in place and exposed to a wide x-ray field emerging from a thick patient.

Moving Grids

The image of grid strips in a radiograph may be distracting to the observer. Also, grid strip shadows sometimes interfere with the identification of small structures such as blood vessels and trabeculae of bone. In the early 1900s, Bucky and Potter developed the moving grid, referred to as the *Potter–Bucky diaphragm*, which removes the distracting image of grid strips by blurring their image across the film. Early Potter–Bucky diaphragms moved in one direction only. Modern moving grids use a reciprocating motion, with the grid making several transits back and forth during an exposure. The linear distance over which the grid moves is small (1 to 5 cm) and permits the use of a focused grid. The motion of the grid must not be parallel to the grid strips and must be rapid enough to move the image of a number of strips across each location in the film during exposure. The motion of the grid must be adjusted to prevent synchronization between the position of the grid strips and the rate of pulsation of the x-ray beam. The direction of motion of the grid changes very rapidly at the limits of grid travel, and the "dwell time" of the grid at these limits is insignificant.

Orthogonal crossed grids are not often used as moving grids. Rhombic crossed grids used in "super-speed recipromatic diaphragms" provide excellent removal of scattered radiation with no image of the grid strips in the radiograph. The travel of the grid is very short and does not cause significant off-center cutoff (see below). Most stationary radiographic tables contain a recipromatic Potter–Bucky diaphragm. However, the development of grids with many strips per inch has reduced the need for moving grids. The maintenance cost for reciprocating grids and their poor efficiency for removing the image of grid strips during short exposure times enhance the attractiveness of stationary grids that contain many strips per inch.

The choice of a radiographic grid for a particular examination depends on factors such as the amount of primary and scattered radiation emerging from the patient, the energy of x rays in the x-ray beam, and the variety of radiographic techniques provided by the x-ray generator.[8]

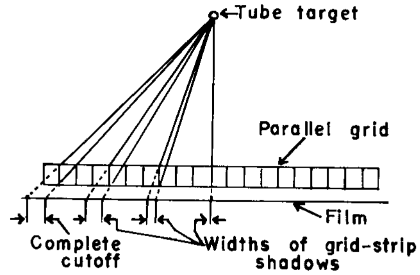
Grid Cutoff

The expression "grid cutoff" is used to describe the loss of primary radiation caused by improper alignment of a radiographic grid. With a parallel grid, cutoff occurs near the edges of a large field because grid strips intercept many primary photons along the edges of the x-ray beam. The width of the shadow of grid strips in a parallel grid increases with distance from the center of the grid (Margin Figure 13-21).

The use of a focused grid at an incorrect target–grid distance also causes grid cutoff. This effect is termed *axial decentering* or *off-distance cutoff* and is depicted in Margin Figure 13-22. The optical density of a radiograph with off-distance cutoff decreases from the center of the radiograph outward. The variation in optical density increases with displacement of the grid from the correct target–grid distance. However, the effect is not objectionable until the displacement exceeds the target–grid distance limits established for the grid. The limits for the target–grid distance are narrow for grids with a high grid ratio and wider for grids with a smaller ratio. The effects of off-distance cutoff are more severe when the target–grid distance is shorter than that recommended and less severe when the target–grid distance is greater.

Lateral decentering or *off-center cutoff* occurs when x rays parallel to the strips of a focused grid converge at a location that is displaced laterally from the target of the x-ray tube (Margin Figure 13-23). *Off-level cutoff* results from tilting of the grid (Margin Figure 13-24). Both off-center cutoff and off-level cutoff cause an overall reduction in optical density across the radiograph. The importance of correct alignment of the x-ray tube, grid, and film increases with the grid ratio.

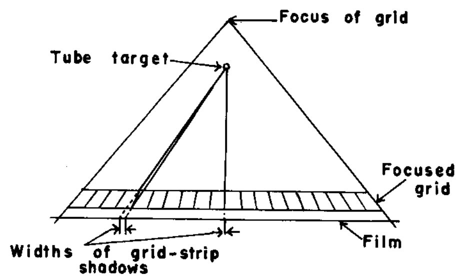
To x-ray photons traveling from an x-ray tube through the patient to an image receptor, a moving grid does not appear to be moving very much at all. Therefore, the function of the grid, to allow photons that are traveling in a straight line from the focal spot to reach the image receptor, is preserved, even though the final image of the grid is blurred. A typical moving grid travels at a maximum speed of 15 cm/sec. It takes x-ray photons only 3 nanoseconds to travel the 100-cm distance from the x-ray tube to the image receptor. During this time, a grid moving at 15 cm/sec will have traveled only 0.5 nanometers—the width of 5 atoms.



MARGIN FIGURE 13-21
Cutoff of primary x-ray photons with a parallel grid.

Air Gap

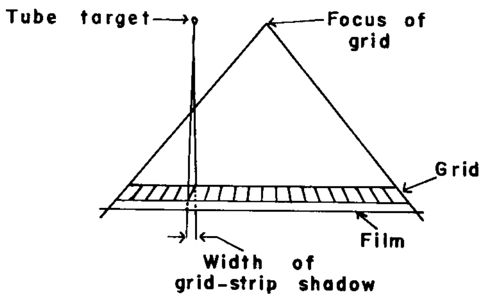
The amount of scattered radiation reaching an x-ray film or film-screen combination may be reduced somewhat by increasing the distance between the patient and the film. This procedure is referred to as an air grid or air gap. The use of an air gap is accompanied by magnification of the image and an increase in geometric unsharpness (see Chapter 16).



MARGIN FIGURE 13-22
Off-distance cutoff with a focused grid.

Moving-Slit Radiography

More efficient techniques for removal of scattered radiation have been sought for many years. One technique that has been proposed periodically is moving-slit radiography, in which one or more slits in an otherwise x-ray-opaque shield move above the patient in synchrony with an equal number of slits in a shield between the patient and the film (Margin Figure 13-25). The long, narrow x-ray beams emerging through the upper slits are scanned across the patient and transmitted to the film through the lower slits below the patient. Radiation scattered by the patient is intercepted by the opaque shield below the patient and does not reach the film. The principal disadvantage of moving-slit radiography is the possibility of image distortion due to motion during the time required for the x-ray beam(s) to scan across the patient. This disadvantage is less cumbersome with newer, "fast" x-ray systems using high-milliamperage generators and rare-earth intensifying screens.[10]

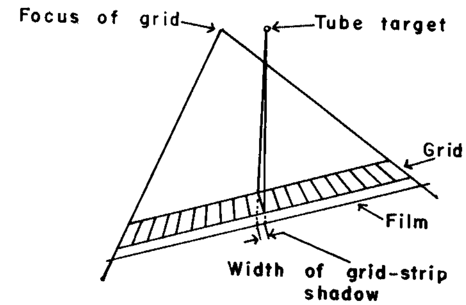


MARGIN FIGURE 13-23
Off-center cutoff with a focused grid.

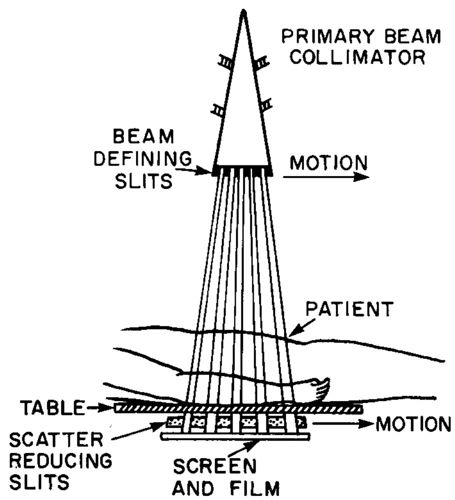
MAGNIFICATION RADIOGRAPHY

If x rays are assumed to originate from a single point on the target of the x-ray tube, then the magnification of a radiographic image is

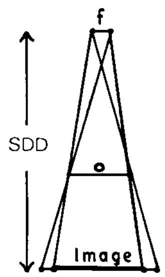
$$\text{Magnification} = \frac{\text{Image size}}{\text{Object size}} = \frac{\text{Target} - \text{film distance}}{\text{Target} - \text{object distance}}$$



MARGIN FIGURE 13-24
Off-level cutoff with a focused grid.



MARGIN FIGURE 13-25
Moving-slit approach to rejection of scattered radiation. (From Hendee, W., Chaney, E., and Rossi, R. *Radiologic Physics, Equipment and Quality Control*. St. Louis, Mosby-Year Book, 1977. Used with permission.)



MARGIN FIGURE 13-26
Principle of enlargement radiography: f is the apparent focal spot of the x-ray tube, o is the object size, and SDD represents the distance between the image receptor and the target of the x-ray tube.

If the distance between the target of the x-ray tube and the x-ray film is constant, then the ratio of image to object size may be increased by moving the object toward the x-ray tube. This method of image enlargement is referred to as *object-shift enlargement* (Margin Figure 13-26). If the distance between the target and object is constant, then the ratio of image size to object size may be increased by moving the film farther from the object. This technique of image enlargement is referred to as *image-shift enlargement*.

The amount of enlargement possible without significant loss of image detail should be determined when radiologic images are magnified. With the image receptor close to the patient, the visibility of image detail is usually determined primarily by the unsharpness of the intensifying screen. As the distance between the screen and the patient is increased by object-shift or image-shift enlargement, no change occurs in the contribution of screen unsharpness to the visibility of image detail. However, the contribution of geometric unsharpness increases steadily with increasing distance between the patient and image receptor. At some patient–receptor distance, geometric unsharpness equals screen unsharpness. Beyond this distance, the visibility of image detail deteriorates steadily as geometric unsharpness increasingly dominates image unsharpness. As a general rule, the image should not be magnified beyond the point at which geometric and screen unsharpness are about equal.

An image also may be magnified by optical enlargement of a processed radiograph. This technique may enhance the visibility of detail but offers no improvement in intrinsic resolution of the image. For most procedures, magnification radiographs obtained directly by geometric enlargement of the image are preferred over those obtained by optical enlargement.[11,12]

DIGITAL RADIOGRAPHY

A number of techniques exist to acquire radiographic images digitally or to convert images acquired in analog fashion into digital format. Of course, it may be argued that even film screen systems are not truly analog because the grain size in film and crystal size in fluorescent screens are examples of the fundamental discreetness of all image receptors and indeed all matter. However, structures such as film grain do not limit current imaging technology, whereas the resolution of currently available display systems such as cathode-ray tubes and the limitations of transmission time and data storage for the massive amounts of information present in images are limitations. Furthermore, the discrete components of "analog" image acquisition devices (e.g., film grains) cannot be manipulated or adjusted individually. A number of interesting techniques for accessing discrete components of images have recently moved from research into clinical usage.

Discrete Digital Detectors

The use of an array of detectors is routine for ultrasound and computed tomography (CT). The exquisite information density of film obviated the need for such devices in radiography until imaging demands that required digital techniques were identified. These demands include increased dynamic range, simultaneous viewing at multiple locations, and the use of digital image processing and enhancement. These newer demands have recently begun to be matched by technologic capabilities. A number of discrete detectors that produce electronic signals that are easily digitized have been used, including fluorescent crystals such as sodium iodide and bismuth germanate, photodiodes, and various semiconductor devices. The idea of individual detectors or groups of detectors intercepting narrowly collimated x-ray beams for the purpose of reducing the influence of scattered photons on image formation has been exploited in the form of line (slit) scanning and point scanning systems. The problem of dynamic range has been addressed in the technique of scanning equalization radiography.[13,14] This system incorporates a feedback mechanism in which the magnitude of signals from the detector causes automatic adjustment of the output of the x-ray tube. The adjustment of tube output in a scanning equalization radiography system reduces the range of exposure rates over which the detector must respond.

Storage Phosphor

Another approach to digital imaging is acquisition of the image on a continuous medium that is specially designed to be digitized. In storage phosphor technology (also known as computed radiography [CR]) the image is acquired on a plate containing crystals of a photostimulatable phosphor. A material such as barium fluorobromide (BaFBr) is capable of storing the energy from an x-ray exposure. When exposed to a strong light source of the appropriate wavelength, the photostimulatable phosphor re-emits the energy as visible light that can be detected by a photomultiplier tube. Thus the storage phosphor plate records a latent image that may be read out some time after x-ray exposure. The readout may be accomplished with a well-collimated intense light source, such as a helium–neon laser beam, so that the size of the stimulated region of the plate remains small to yield good resolution. The electrical signals from the photomultiplier or other light-sensing device are digitized with an analog-to-digital converter. Once the image is stored in digital form, it may be viewed on a high-resolution monitor or printed out on film.

One of the advantages of a stimulatable phosphor plate system over conventional film is an improvement in dynamic range. Radiographic film operates over possible exposures ranging from the exposure required to reach the shoulder of the characteristic curve to that required to rise above the toe. This range typically causes exposure differences of a factor of approximately 100. The dynamic range of a storage phosphor is on the order of 10,000. Thus the storage phosphor has greater latitude (i.e., is more "forgiving") if an incorrect exposure is used. To preserve the information contained within such a broad dynamic range, the analog-to-digital converter must have a sufficient bit depth (see Chapter 10). A typical storage phosphor readout system is capable of digitizing a 2000×2000 pixel matrix or greater, with each pixel taking one of $2^{10} = 1024$ possible values. The images may then be processed according to a number of possible gray-scale mapping functions or simply windowed and leveled in a fashion similar to that used for magnetic resonance and CT images. Image processing techniques may also be used to enhance the appearance of edges of structures such as tubes in chest radiographs taken in intensive care units.[15]

Some of the advantages of storage phosphor or CR systems over conventional film-screen approaches include the potential for reduction of patient exposure and the virtual elimination of retakes due to improper technique. There is some advantage of the storage phosphor system over other digital radiographic techniques in that the storage phosphor plate simply replaces the screen/film cassette and does not represent a significant change in patient positioning or other imaging procedures.

Example 13-6

Calculate the spatial resolution of a CR system in which a 14×17-in. storage phosphor plate is digitized at a matrix size of 2000×2510.

$$\text{Resolution element or pixel size} = r_{14} = \frac{14 \text{ in.}}{2000} = 0.007 \text{ in.} = 0.18 \text{ mm}$$

$$= r_{17} = \frac{17 \text{ in.}}{2510} = 0.007 \text{ in.} = 0.18 \text{ mm}$$

Thus the resolution element size is the same in both the 14- and 17-in. dimensions. The number of line pairs per millimeter that could be displayed equals the number of pixel pairs per millimeter.

$$\frac{\text{Line pairs}}{\text{millimeter}} = \frac{1}{2(r)} = \frac{1}{2(0.18 \text{ mm})} = 2.8 \frac{\text{Line pairs}}{\text{millimeter}}$$

Note that the spatial resolution of the current generation of storage phosphor systems is less than the six to ten line pairs per millimeter available with film-screen systems.

Film Scanning

Radiographic films may be digitized after they are acquired with conventional film/screen systems. In a typical film digitizer, a laser beam is scanned across a film. The pattern of optical densities on the film modulates the transmitted light. A light detector on the opposite side of the film converts the transmitted laser light to an electrical signal that is digitized by an analog-to-digital converter.[16] Spatial resolution of a film scanner is determined by spot size, defined as the size of the laser beam as it strikes the film. Spot sizes down to 50 μm (50×10^{-6} m) are available, but most clinical systems use 100 μm or larger. Because the laser spot must be scanned across the entire area of the image, a decrease in the diameter of the spot size by a factor of 2 requires a scan time that is 4 × longer, if no other adjustments are made in the system. Commercially available laser film scanners have matrix sizes of at least 2000 × 2000 × 10 or 12 bits.

Example 13-7

Find the total number of bytes stored for a film scanning system with a 2048 × 2048 matrix if an 8-bit analog-to-digital converter is used.

$$\begin{aligned}\text{Matrix size} &= 2048 \times 2048 \times 8 \text{ bits}\\ &\simeq 33{,}570{,}000 \text{ bits}\end{aligned}$$

However, 8 bits = 1 byte, and

$$\begin{aligned}\text{Matrix size} &= 33{,}570{,}000 \text{ bits} \div 8 \text{ bits/byte}\\ &\simeq 4{,}200{,}000 \text{ bytes}\\ &\simeq 4.2 \text{ Mbytes}\end{aligned}$$

Each image requires 4 Mbytes.

A film scanner has the obvious advantage over other digital radiographic techniques in that it does not disrupt routine imaging procedures. All procedures (e.g., technique, patient positioning, film development) are the same up to the point of appearance of the developed radiograph. Compared with other digital techniques, disadvantages of a film scanner include its inability to correct for large variations in film density or gray-scale mapping, the need for handling individual sheets of film, and the time required for scanning.

Large-Area Digital Image Receptors for Radiography

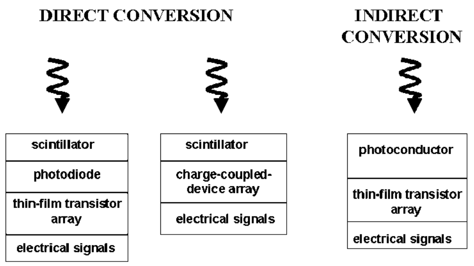
Large-area (e.g., 35 cm × 43 cm) digital detectors have been under development for over 30 years, beginning with the first charge-coupled devices (CCDs) developed at Bell Laboratories in 1969.[17] CCDs convert visible light to electrical signals. Digital detection has revolutionized light and video camera technology, and various digital technologies are ubiquitous in photography and video recording today. However, the extreme demands of resolution, low noise, and high dynamic range required in medical radiography present major technological challenges to application of digital detectors in radiography. Over the past two decades, the promise of digital detectors has remained just out of reach in terms of both performance and cost effectiveness. There has been great incentive to continue development, however, because using detectors that can be read electronically in a matter of seconds eliminates the need for trips to a processor (as is required with film and/or a charged-plate reader used for computed radiography). Also, the possibility of rapid image processing—even just window and level manipulation—yields an overall increase in the quality of the final images.

Only recently have advances in the technology of fabrication and manufacture allowed the production of large-area flat panel detectors that equal, or even surpass (at least in theory), the performance of film-screen and computed radiography systems. The most important feature of flat panel detectors in this comparison is the higher quantum efficiency for detection and conversion of x rays into digital signals with acceptably low noise over a large exposure range.[18] The two basic types of large-area flat panel x-ray detection systems, *direct conversion* and *indirect conversion,* are shown in Margin Figure 13-27.

The objective of a digital detector in radiography is to convert the energy of x-ray photons into electrical signals that signify the magnitude of the exposure in different pixels. In direct capture systems, the energy of the x rays is converted to an electrical signal in a single layer of a material called a *photoconductor*. Various materials can be used to construct photoconductors, including amorphous selenium, cadmium zinc telluride, and lead iodide.[19] When x rays strike a photoconductor conversion plate, electron–hole pairs are created. Electrical fields are applied between the front and back surfaces of the photoconductor to force separation of the electron–hole pairs and their transfer to one of the charged surfaces. Each surface is a thin-film transistor (TFT) array, a grid of transistors that can be interrogated electronically ("read out") to determine the amount of electrical charge present above each transistor. The TFT array corresponds to an array of pixels in the final image, with the number of transistors equaling the number of pixels. Flat-panel detectors having pixel sizes on the order of 200 micrometers (2.5 lp/mm resolution) are available for chest imaging, and a new generation of detectors with spatial resolution in the 50- to 100-micrometer (5- to 10-lp/mm) range is becoming available for mammography.

In indirect conversion systems, a scintillation material converts the energy of the x-ray photons to visible light. Scintillation materials used for this purpose include amorphous silicon and cesium iodide. The pattern of visible light produced by the scintillation conversion plate can be read out by a photoconductor fabricated to convert visible light to electrical signals. Photoconductors used in indirect conversion systems include CCDs and materials such as amorphous silicon coupled with a TFT array.

An important feature of the newer generation of flat-panel detectors is their higher detective quantum efficiency (DQE). DQE is a measure of the detection efficiency that includes the effects of noise as well as resolution at various spatial frequencies. The DQE of current flat-panel digital detectors is over 65%,[20] while that of computed radiography storage phosphor systems is approximately 35%,[21] and that of screen-film systems is approximately 25%.[22] Because of the higher DQE, flat-panel detectors are expected to use radiation dose more efficiently, and they could theoretically provide lower noise images over the range of spatial frequencies encountered in medical imaging.



MARGIN FIGURE 13-27
Some large-area detectors provide indirect conversion of x-ray energy to electrical charge through intermediate steps involving photodiodes or charge-coupled-devices. Other large-area detectors provide direct conversion of x-ray energy to electrical charge through the use of a photoconductor.

PROBLEMS

*13-1. Adjacent regions of a radiograph have optical densities of 1.0 and 1.5. What is the difference in the transmission of light through the two regions?

*13–2. Optical densities of 1.0 and 1.5 are measured for adjacent regions of a radiograph obtained with x-ray film with a film gamma of 1.0. If the exposure was 40 mR to the region with an optical density of 1.0, what was the exposure to the region with an optical density of 1.5?

13–3. Explain why poor contact between an intensifying screen and an x-ray film reduces the resolution of the radiographic image.

13–4. The effective atomic number is greater for barium lead sulfate than for calcium sulfate or zinc sulfide. Explain why intensifying screens of barium lead sulfate are particularly useful for radiography at higher tube voltages.

13–5. Explain the meaning of the expressions "grid cutoff," "off-distance cutoff," "off-center cutoff," and "off-level cutoff."

13–6. Explain why a crossed grid provides a greater contrast improvement factor than a linear grid with the same grid ratio.

13–7. Discuss the relationships between average gradient, film contrast, and latitude.

13–8. Define film speed and discuss its significance.

13–9. Describe the principle of moving-slit radiography.

13–10. Describe two reasons why rare-earth screens are faster than conventional calcium tungstate screens.

13–11. Discuss the principle of xeroradiography and ionic radiography.

*For those problems marked with an asterisk, answers are provided on page 492.

SUMMARY

- Radiographic film consists of a silver halide emulsion on a support base with a protective coating.
- $\mathrm{OD} = \log \frac{1}{T}$
- Latitude is the exposure range over which acceptable optical densities are obtained.
- Average gradient is the slope of the H and D curve in the region where it is linear.
- Gamma is the maximum slope of the H and D curve.
- Speed is the inverse of the exposure required to achieve an optical density of 1.0 above base + fog.
- Grid ratio is the ratio of the height of the grid strips to the width of the spacing between strips.
- The contrast improvement factor of a grid is the ratio of the contrast with the grid to the contrast without the grid.
- Selectivity of a grid is the ratio of primary radiation to scattered radiation that is transmitted by the grid.
- Storage (photostimulable) phosphor image receptors record an image and are then interrogated with a laser beam to "read out" the image.
- Large-area digital detectors may be classified as direct conversion or indirect conversion. Indirect conversion detectors first convert the energy of photons to visible light before recording a signal with a light sensitive digital detector.

REFERENCES

1. Ter-Pogossian, M. *The Physical Aspects of Diagnostic Radiology*. New York, Harper & Row, 1967.
2. Arnold, B., Eisenberg, H., and Bjärngard, B. Measurements of reciprocity law failure in green-sensitive x-ray films. *Radiology* 1978; **126:**493.
3. Seemann, H. *Physical and Photographic Principles of Medical Radiography*. New York, John Wiley & Sons, 1968, pp. 69.
4. Coltman, J., Ebbighauser, E., and Altar, W. Physical properties of calcium tungstate x-ray screens. *J. Appl. Phys.* 1947; **18:**530.
5. Towers, S. X-ray intensifying screens: Part I. *X-Ray Focus* 1967; **8:**2.
6. Rossi, R., Hendee, W., and Ahrens, C. An evaluation of rare earth screen/film combinations. *Radiology* 1976; **121:**465.
7. Hondius-Boldingh, W. *Grids to Reduce Scattered X-Rays in Medical Radiography,* Philips Research Reports Supplement 1. Eindhoven, Netherlands, Philips Research Laboratories, 1964.
8. *Characteristics and Applications of X-Ray Grids*. Cincinnati, Liebel-Flarsheim, 1968.
9. International Commission on Radiological Units and Measurements. *Methods of Evaluating Radiological Equipment and Materials*. Recommendations of the ICRU, Report 10f. National Bureau of Standards Handbook 89, 1963.
10. Barnes, G., Cleare, H., and Brezovich, I. Reduction of scatter in diagnostic radiology by means of a scanning multiple slit assembly. *Radiology* 1976; **120:**691.
11. Enlargement radiography, a review of 0.3 mm techniques with bibliography. *Cathode Press* 1967; **24:**32.
12. Isard, H., et al. *An Atlas of Serial Magnification Roentgenography,* Cathode Press Supplement. Stamford, CT, Machlett Laboratories, 1968.
13. Wandtke, J. C., and Plewes, D. B. Improved chest disease detection with scanning equalization radiography. *AJR* 1985; **145:**979–983.
14. Korhola, O., Riihimaki, E., and Savikurki, S. Reduction of radiation scatter with a multiple pencil-beam imaging device. *Radiology* 1987; **165:**257–259.
15. Merritt, C. R. B., Tutton, R. H., Bell, K. A., et al. Clinical application of digital radiography: Computed radiographic imaging. *Radiographics* 1985; **5:**397–414.
16. Anderson, W. F. A Laser multiformat imager for medical applications, in Schneider, R. H., Dwyer, S. J. II (eds.), *Medical Imaging. Proceedings of the Society of Photo Optical Instrumentation Engineers (SPIE),* Vol. 767, 1987, pp. 516–523.
17. Boyle, W., and Smith, G. Charge coupled semiconductor devices. *Bell Systems Tech. J.* April 1970; **49:**587.
18. Chotas, M. S., Dobbins, J. T., Ravin, C. E. Principles of digital radiography with large-area, electronically readable detectors: A review of the basics. *Radiology* 1999, **210:**595–599.
19. Pisano, E. D., et al. Current status of full-field digital mammography. *Acta Radiol.* 2000; **7:**266–280.
20. Floyd, C. E., et al. Imaging characteristics of an amorphous silicon flat-panel detector for digital chest radiography. *Radiology* 2001; 218(3):683–688.
21. Dobbins, J. T., et al. DQE(f) of four generations of computed radiography acquisition devices. *Med. Phys.* 1995; **22:**1581–1593.
22. Bunch, P. C. Performance characteristics of asymmetric zero-crossover screen-film systems. *Proc. SPIE* 1992; **1653:**46–65.